Compute Cell Scheduling Records

This document records how tasks were scheduled on small compute clusters. The workload and the machine failures are real records from a production compute cell of about 12,500 machines (the Google cluster trace of May 2011, clusterdata-2011-2, CC-BY 4.0): every task's submission time, job, priority, CPU and memory request and run time, and every machine's capacity, failure time and return time, are taken from that trace (ids are renamed). Where and when each task ran in the small cluster is not recorded in the trace: it is computed from those records by the scheduling policy stated below, applied exactly.

The document is divided into chapters titled 'Cell window <NN>'. Each chapter is an independent run of one small cluster over a window of 1200 seconds. Questions refer to a chapter by that title. Times are whole seconds from the start of the window (time 0); the window covers times 0 to 1200.

Each chapter has four sections. 'Machines' lists the cluster's machines with their CPU and memory capacity, the time a machine failed (if it did) and the time it returned (if it did). 'Task log' lists, in time order, every event of every task from time 0 to the window end: submit, start, finish, evict and fail, with the task's job, priority, CPU and memory request, run time, and the machine involved. 'Events' lists machine failures and returns, evictions and restarts, and the longest waits. 'Queue at window end' gives the state after the events at time 1200: the tasks still running (with their remaining run time) and the tasks still waiting, in queue order.

Tasks. A task belongs to a job and has a priority from 0 to 11 (higher is more important). Priorities 9 to 11 are the *production* tier and priorities 0 and 1 the *free* tier. Each task requests an amount of CPU and of memory, which it holds on one machine while it runs, and has a run time: once started it runs for exactly that many seconds and then finishes, unless it is stopped first. The run time is given on every row of the task log, starting with the task's submit row.

Scheduling policy. Every window starts with all machines empty and no task waiting. At every time at which something happens, in this order:

1. **Finishes.** Tasks whose run time is over finish and free their CPU and memory.

2. **Failures.** Machines that fail at this time go down. Each task running on such a machine stops (a *fail* event) and goes back to the queue.

3. **Returns.** Machines that return at this time come back, empty, with their capacity.

4. **Submissions.** Tasks submitted at this time join the queue.

5. **Scheduling round.** The waiting tasks are taken one at a time in *queue order*: higher priority first, then earlier submission time, then lower task id. Each task, in turn:

a. **First fit.** The task starts on the first machine, in machine id order, that is up and whose free CPU and free memory are both at least the task's requests (subject to the reservation rule in c).

b. **Preemption.** If it fits nowhere, no task holds a reservation yet in this round, and it is a production task, it may evict free-tier tasks from one machine. It looks at the up machines in id order. On each it considers the free-tier tasks running there, lowest priority first, then the most recently started first (then the higher task id first), and evicts them one by one in that order until it fits; it takes the first machine where this works, evicting only as many as needed there, and starts on it. If no machine works, nothing is evicted. An evicted task goes back to the queue at once (it may start again later in the same round); it loses its progress and will run its full run time again.

c. **Reservation and backfilling.** The first task in the round that can start neither by a. nor by b. gets a *reservation*: the earliest time at which it could start, computed from the run times of the tasks now running (assuming no new submissions, failures or returns), on the first machine, in id order, where that earliest time is reached. Every later task in the round may still start by first fit, but on the reserved machine only if it will finish no later than the reserved time, or if its requests fit in what the reserved machine will still have free at the reserved time after the reserved task's requests (and after other such starts on it in the same round). On the other machines it starts as usual. Later tasks do not preempt. A task that would not fit on any up machine even if that machine were empty gets no reservation and simply keeps waiting; the next task that cannot start may then get the reservation.

Tasks stopped by an eviction or a failure keep their original submission time (and so their place in queue order) and start again from the beginning. A task never moves while it runs.

Order of rows. Task log rows with the same time follow the steps above: finishes (by task id), failures (by machine, then task id), submissions (by task id), then the scheduling round's starts and evictions in the order the round makes them (each eviction just before the start that caused it).

Terms used in the Events section and in questions. A task's *wait* is the time from its submission to its first start. A task is *restarted* when it starts again after an eviction or a failure. The number of *evictions* counts evict events (one per evicted task). 'Running on a machine at time t' means running after all the events at time t.

The cluster and the workload. Each window has 8 machines, ids M01 to M08. Each is modelled on a real machine of the cell (one of them is a machine that failed during the window in the trace) and has 1/8 of that machine's recorded CPU and memory capacity. A failed machine returns at the time the trace records it coming back, if that is within 6 hours; otherwise it does not return. The tasks are a sample of the real tasks submitted in the cell during the window, chosen by this fixed rule: jobs submitted in the window are taken in a random order, first up to 6 production jobs, then free-tier jobs until the offered load reaches 54%, then other jobs until it reaches 90% (a job that would push it above 95% is skipped). A job contributes its first 4 tasks (16 for a free-tier job) that finished normally in the trace and fit on at least one machine of the window. Offered load = the larger, over CPU and memory, of the sum of request x run time inside the window over all tasks, divided by the cluster's total capacity x the window length.

After the window. For questions about what happens after time 1200: no new task is submitted and no machine fails after the window; a machine that is down returns at its listed return time (if any); the policy continues unchanged from the state given in 'Queue at window end'.

Units. Times and run times in seconds. CPU and memory in units of 1/100,000 of the largest machine in the original cell (the trace's normalised values, rounded to whole units).

Chapters in this document (itops_d05): Cell window 05, Cell window 12, Cell window 13, Cell window 15, Cell window 16.

Cell window 05

Times 0 to 1200. The cluster has 8 machines. 73 tasks of 24 jobs are submitted during the window.

Cell window 05: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	512	597
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	12500	12500	-	-
M05	6250	3116	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	6250	6244	-	-

Cell window 05: Task log

time	task	job	priority	CPU	memory	run time	event	machine
118	T001	J01	8	6250	3180	1274	submit	-
118	T001	J01	8	6250	3180	1274	start	M01
141	T002	J02	4	1250	466	201	submit	-
141	T003	J03	8	937	777	393	submit	-
141	T003	J03	8	937	777	393	start	M02
141	T002	J02	4	1250	466	201	start	M02
144	T004	J04	9	1250	1590	38	submit	-
144	T004	J04	9	1250	1590	38	start	M02
182	T004	J04	9	1250	1590	38	finish	M02
183	T005	J05	8	937	466	103	submit	-
183	T005	J05	8	937	466	103	start	M02
184	T006	J06	9	62	466	2464	submit	-
184	T006	J06	9	62	466	2464	start	M02
230	T007	J07	9	1268	466	2346	submit	-
230	T007	J07	9	1268	466	2346	start	M02
232	T008	J08	1	687	932	2343	submit	-
232	T009	J08	1	687	932	2340	submit	-
232	T010	J09	0	687	932	2277	submit	-
232	T011	J09	0	687	932	2287	submit	-
232	T012	J09	0	687	932	2285	submit	-
232	T013	J09	0	687	932	2288	submit	-
232	T014	J09	0	687	932	2338	submit	-
232	T015	J09	0	687	932	2280	submit	-
232	T016	J09	0	687	932	2299	submit	-
232	T017	J09	0	687	932	2288	submit	-
232	T018	J09	0	687	932	2282	submit	-
232	T019	J09	0	687	932	2280	submit	-
232	T020	J09	0	687	932	2308	submit	-
232	T021	J09	0	687	932	2295	submit	-
232	T022	J09	0	687	932	2277	submit	-
232	T023	J09	0	687	932	2298	submit	-
232	T024	J09	0	687	932	2315	submit	-
232	T025	J09	0	687	932	2296	submit	-
232	T008	J08	1	687	932	2343	start	M02
232	T009	J08	1	687	932	2340	start	M02
232	T010	J09	0	687	932	2277	start	M03
232	T011	J09	0	687	932	2287	start	M03
232	T012	J09	0	687	932	2285	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
232	T013	J09	0	687	932	2288	start	M03
232	T014	J09	0	687	932	2338	start	M03
232	T015	J09	0	687	932	2280	start	M03
232	T016	J09	0	687	932	2299	start	M04
232	T017	J09	0	687	932	2288	start	M04
232	T018	J09	0	687	932	2282	start	M04
232	T019	J09	0	687	932	2280	start	M04
232	T020	J09	0	687	932	2308	start	M04
232	T021	J09	0	687	932	2295	start	M04
232	T022	J09	0	687	932	2277	start	M04
232	T023	J09	0	687	932	2298	start	M04
232	T024	J09	0	687	932	2315	start	M04
232	T025	J09	0	687	932	2296	start	M04
255	T026	J10	9	6250	3180	309	submit	-
255	T026	J10	9	6250	3180	309	start	M06
286	T005	J05	8	937	466	103	finish	M02
311	T027	J11	0	6873	1193	3496	submit	-
311	T028	J11	0	6873	1193	3499	submit	-
311	T029	J11	0	6873	1193	3499	submit	-
311	T030	J11	0	6873	1193	3501	submit	-
342	T002	J02	4	1250	466	201	finish	M02
348	T031	J12	8	1250	777	54	submit	-
348	T032	J13	8	625	155	97	submit	-
348	T031	J12	8	1250	777	54	start	M02
348	T032	J13	8	625	155	97	start	M02
367	T033	J14	8	1250	777	66	submit	-
367	T033	J14	8	1250	777	66	start	M04
375	T034	J15	0	6250	1244	52	submit	-
375	T034	J15	0	6250	1244	52	start	M05
402	T031	J12	8	1250	777	54	finish	M02
427	T034	J15	0	6250	1244	52	finish	M05
433	T033	J14	8	1250	777	66	finish	M04
445	T032	J13	8	625	155	97	finish	M02
512	T001	J01	8	6250	3180	1274	fail	M01
512	T001	J01	8	6250	3180	1274	start	M08
534	T003	J03	8	937	777	393	finish	M02
550	T035	J16	8	6250	1590	150	submit	-
550	T035	J16	8	6250	1590	150	start	M05
552	T036	J17	4	6250	1590	149	submit	-
552	T036	J17	4	6250	1590	149	start	M07
564	T026	J10	9	6250	3180	309	finish	M06
583	T037	J18	4	3125	1125	47	submit	-
583	T037	J18	4	3125	1125	47	start	M02
615	T038	J19	9	4688	6360	783	submit	-
615	T025	J09	0	687	932	2296	evict	M04
615	T024	J09	0	687	932	2315	evict	M04
615	T023	J09	0	687	932	2298	evict	M04
615	T022	J09	0	687	932	2277	evict	M04
615	T038	J19	9	4688	6360	783	start	M04
615	T022	J09	0	687	932	2277	start	M01
615	T023	J09	0	687	932	2298	start	M01
615	T024	J09	0	687	932	2315	start	M01
615	T025	J09	0	687	932	2296	start	M01
630	T037	J18	4	3125	1125	47	finish	M02
700	T035	J16	8	6250	1590	150	finish	M05
701	T036	J17	4	6250	1590	149	finish	M07
708	T039	J20	1	769	466	312	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
708	T039	J20	1	769	466	312	start	M01
722	T040	J21	0	625	155	294	submit	-
722	T041	J21	0	625	155	314	submit	-
722	T042	J21	0	625	155	296	submit	-
722	T043	J21	0	625	155	306	submit	-
722	T044	J21	0	625	155	304	submit	-
722	T045	J21	0	625	155	306	submit	-
722	T046	J21	0	625	155	309	submit	-
722	T047	J21	0	625	155	285	submit	-
722	T048	J21	0	625	155	289	submit	-
722	T049	J21	0	625	155	290	submit	-
722	T050	J21	0	625	155	296	submit	-
722	T051	J21	0	625	155	298	submit	-
722	T052	J21	0	625	155	288	submit	-
722	T053	J21	0	625	155	288	submit	-
722	T054	J21	0	625	155	287	submit	-
722	T055	J21	0	625	155	286	submit	-
722	T040	J21	0	625	155	294	start	M01
722	T041	J21	0	625	155	314	start	M01
722	T042	J21	0	625	155	296	start	M01
722	T043	J21	0	625	155	306	start	M01
722	T044	J21	0	625	155	304	start	M02
722	T045	J21	0	625	155	306	start	M02
722	T046	J21	0	625	155	309	start	M02
722	T047	J21	0	625	155	285	start	M02
722	T048	J21	0	625	155	289	start	M02
722	T049	J21	0	625	155	290	start	M03
722	T050	J21	0	625	155	296	start	M03
722	T051	J21	0	625	155	298	start	M03
722	T052	J21	0	625	155	288	start	M04
722	T053	J21	0	625	155	288	start	M04
722	T054	J21	0	625	155	287	start	M04
722	T055	J21	0	625	155	286	start	M05
856	T056	J22	8	6873	9546	549	submit	-
923	T057	J23	0	625	219	261	submit	-
923	T058	J23	0	625	219	260	submit	-
923	T059	J23	0	625	219	258	submit	-
923	T060	J23	0	625	219	258	submit	-
923	T061	J23	0	625	219	259	submit	-
923	T062	J23	0	625	219	258	submit	-
923	T063	J23	0	625	219	257	submit	-
923	T064	J23	0	625	219	257	submit	-
923	T065	J23	0	625	219	257	submit	-
923	T066	J23	0	625	219	257	submit	-
923	T067	J23	0	625	219	260	submit	-
923	T068	J23	0	625	219	257	submit	-
923	T069	J23	0	625	219	260	submit	-
923	T070	J23	0	625	219	259	submit	-
923	T071	J23	0	625	219	259	submit	-
923	T072	J23	0	625	219	261	submit	-
923	T057	J23	0	625	219	261	start	M05
923	T058	J23	0	625	219	260	start	M05
923	T059	J23	0	625	219	258	start	M05
923	T060	J23	0	625	219	258	start	M05
923	T061	J23	0	625	219	259	start	M05
923	T062	J23	0	625	219	258	start	M05
923	T063	J23	0	625	219	257	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
923	T064	J23	0	625	219	257	start	M05
923	T065	J23	0	625	219	257	start	M05
923	T066	J23	0	625	219	257	start	M06
923	T067	J23	0	625	219	260	start	M06
923	T068	J23	0	625	219	257	start	M06
923	T069	J23	0	625	219	260	start	M06
923	T070	J23	0	625	219	259	start	M06
923	T071	J23	0	625	219	259	start	M06
923	T072	J23	0	625	219	261	start	M06
1007	T047	J21	0	625	155	285	finish	M02
1008	T055	J21	0	625	155	286	finish	M05
1009	T054	J21	0	625	155	287	finish	M04
1010	T052	J21	0	625	155	288	finish	M04
1010	T053	J21	0	625	155	288	finish	M04
1011	T048	J21	0	625	155	289	finish	M02
1012	T049	J21	0	625	155	290	finish	M03
1016	T040	J21	0	625	155	294	finish	M01
1018	T042	J21	0	625	155	296	finish	M01
1018	T050	J21	0	625	155	296	finish	M03
1020	T039	J20	1	769	466	312	finish	M01
1020	T051	J21	0	625	155	298	finish	M03
1026	T044	J21	0	625	155	304	finish	M02
1028	T043	J21	0	625	155	306	finish	M01
1028	T045	J21	0	625	155	306	finish	M02
1031	T046	J21	0	625	155	309	finish	M02
1036	T041	J21	0	625	155	314	finish	M01
1155	T073	J24	9	6250	3180	330	submit	-
1155	T025	J09	0	687	932	2296	evict	M01
1155	T024	J09	0	687	932	2315	evict	M01
1155	T023	J09	0	687	932	2298	evict	M01
1155	T022	J09	0	687	932	2277	evict	M01
1155	T073	J24	9	6250	3180	330	start	M01
1155	T022	J09	0	687	932	2277	start	M02
1155	T023	J09	0	687	932	2298	start	M02
1155	T024	J09	0	687	932	2315	start	M02
1155	T025	J09	0	687	932	2296	start	M06
1180	T063	J23	0	625	219	257	finish	M05
1180	T064	J23	0	625	219	257	finish	M05
1180	T065	J23	0	625	219	257	finish	M05
1180	T066	J23	0	625	219	257	finish	M06
1180	T068	J23	0	625	219	257	finish	M06
1181	T059	J23	0	625	219	258	finish	M05
1181	T060	J23	0	625	219	258	finish	M05
1181	T062	J23	0	625	219	258	finish	M05
1182	T061	J23	0	625	219	259	finish	M05
1182	T070	J23	0	625	219	259	finish	M06
1182	T071	J23	0	625	219	259	finish	M06
1183	T058	J23	0	625	219	260	finish	M05
1183	T067	J23	0	625	219	260	finish	M06
1183	T069	J23	0	625	219	260	finish	M06
1184	T057	J23	0	625	219	261	finish	M05
1184	T072	J23	0	625	219	261	finish	M06

Cell window 05: Events

Time 512: machine M01 failed; task T001 stopped and went back to the queue.

Time 512: task T001 restarted on machine M08.

Time 597: machine M01 returned.

Time 615: task T038 (priority 9) evicted task T022 (priority 0) from machine M04.

Time 615: task T038 (priority 9) evicted task T023 (priority 0) from machine M04.

Time 615: task T038 (priority 9) evicted task T024 (priority 0) from machine M04.

Time 615: task T038 (priority 9) evicted task T025 (priority 0) from machine M04.

Time 615: task T022 restarted on machine M01.

Time 615: task T023 restarted on machine M01.

Time 615: task T024 restarted on machine M01.

Time 615: task T025 restarted on machine M01.

Time 1155: task T073 (priority 9) evicted task T022 (priority 0) from machine M01.

Time 1155: task T073 (priority 9) evicted task T023 (priority 0) from machine M01.

Time 1155: task T073 (priority 9) evicted task T024 (priority 0) from machine M01.

Time 1155: task T073 (priority 9) evicted task T025 (priority 0) from machine M01.

Time 1155: task T022 restarted on machine M02.

Time 1155: task T023 restarted on machine M02.

Time 1155: task T024 restarted on machine M02.

Time 1155: task T025 restarted on machine M06.

Every task started at its submission time.

Cell window 05: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T073	J24	9	6250	3180	M01	1155	285
T006	J06	9	62	466	M02	184	1448
T007	J07	9	1268	466	M02	230	1376
T008	J08	1	687	932	M02	232	1375
T009	J08	1	687	932	M02	232	1372
T022	J09	0	687	932	M02	1155	2232
T023	J09	0	687	932	M02	1155	2253
T024	J09	0	687	932	M02	1155	2270
T010	J09	0	687	932	M03	232	1309
T011	J09	0	687	932	M03	232	1319
T012	J09	0	687	932	M03	232	1317
T013	J09	0	687	932	M03	232	1320
T014	J09	0	687	932	M03	232	1370
T015	J09	0	687	932	M03	232	1312
T016	J09	0	687	932	M04	232	1331
T017	J09	0	687	932	M04	232	1320
T018	J09	0	687	932	M04	232	1314
T019	J09	0	687	932	M04	232	1312
T020	J09	0	687	932	M04	232	1340
T021	J09	0	687	932	M04	232	1327
T038	J19	9	4688	6360	M04	615	198
T025	J09	0	687	932	M06	1155	2251

task	job	priority	CPU	memory	machine	started	remaining
T001	J01	8	6250	3180	M08	512	586

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T056	J22	8	6873	9546	856	549
T027	J11	0	6873	1193	311	3496
T028	J11	0	6873	1193	311	3499
T029	J11	0	6873	1193	311	3499
T030	J11	0	6873	1193	311	3501

All machines are up at time 1200.

Cell window 12

Times 0 to 1200. The cluster has 8 machines. 156 tasks of 66 jobs are submitted during the window.

Cell window 12: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	3116	-	-
M04	6250	6244	449	1453
M05	6250	3116	-	-
M06	6250	6244	-	-
M07	6250	3116	-	-
M08	12500	12500	-	-

Cell window 12: Task log

time	task	job	priority	CPU	memory	run time	event	machine
26	T001	J01	0	736	317	311	submit	-
26	T002	J02	4	1250	466	349	submit	-
26	T002	J02	4	1250	466	349	start	M01
26	T001	J01	0	736	317	311	start	M01
28	T003	J03	0	1250	835	148	submit	-
28	T004	J03	0	1250	835	200	submit	-
28	T005	J03	0	1250	835	218	submit	-
28	T006	J03	0	1250	835	167	submit	-
28	T007	J03	0	1250	835	143	submit	-
28	T008	J03	0	1250	835	175	submit	-
28	T009	J03	0	1250	835	196	submit	-
28	T010	J03	0	1250	835	136	submit	-
28	T011	J03	0	1250	835	199	submit	-
28	T012	J03	0	1250	835	125	submit	-
28	T013	J03	0	1250	835	147	submit	-
28	T014	J03	0	1250	835	217	submit	-
28	T015	J03	0	1250	835	308	submit	-
28	T016	J03	0	1250	835	234	submit	-
28	T017	J03	0	1250	835	144	submit	-
28	T018	J03	0	1250	835	306	submit	-
28	T003	J03	0	1250	835	148	start	M01
28	T004	J03	0	1250	835	200	start	M01
28	T005	J03	0	1250	835	218	start	M02
28	T006	J03	0	1250	835	167	start	M02
28	T007	J03	0	1250	835	143	start	M02
28	T008	J03	0	1250	835	175	start	M02
28	T009	J03	0	1250	835	196	start	M02
28	T010	J03	0	1250	835	136	start	M03
28	T011	J03	0	1250	835	199	start	M03
28	T012	J03	0	1250	835	125	start	M03
28	T013	J03	0	1250	835	147	start	M04
28	T014	J03	0	1250	835	217	start	M04
28	T015	J03	0	1250	835	308	start	M04
28	T016	J03	0	1250	835	234	start	M04
28	T017	J03	0	1250	835	144	start	M04
28	T018	J03	0	1250	835	306	start	M05
31	T019	J04	1	1562	1553	105	submit	-
31	T019	J04	1	1562	1553	105	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
57	T020	J05	4	3125	777	371	submit	-
57	T021	J05	4	3125	777	358	submit	-
57	T022	J05	4	3125	777	363	submit	-
57	T023	J05	4	3125	777	359	submit	-
57	T024	J06	4	625	311	240	submit	-
57	T025	J07	4	625	155	40	submit	-
57	T026	J08	4	3125	1125	35	submit	-
57	T027	J09	8	1874	199	114	submit	-
57	T027	J09	8	1874	199	114	start	M03
57	T020	J05	4	3125	777	371	start	M06
57	T021	J05	4	3125	777	358	start	M06
57	T022	J05	4	3125	777	363	start	M07
57	T023	J05	4	3125	777	359	start	M07
57	T024	J06	4	625	311	240	start	M01
57	T025	J07	4	625	155	40	start	M01
57	T026	J08	4	3125	1125	35	start	M08
74	T028	J10	0	1250	197	49	submit	-
74	T028	J10	0	1250	197	49	start	M05
81	T029	J11	0	1250	197	44	submit	-
81	T029	J11	0	1250	197	44	start	M05
92	T026	J08	4	3125	1125	35	finish	M08
97	T025	J07	4	625	155	40	finish	M01
116	T030	J12	1	1562	1553	116	submit	-
116	T030	J12	1	1562	1553	116	start	M08
123	T028	J10	0	1250	197	49	finish	M05
125	T029	J11	0	1250	197	44	finish	M05
136	T019	J04	1	1562	1553	105	finish	M05
153	T012	J03	0	1250	835	125	finish	M03
154	T031	J13	6	937	1244	84	submit	-
154	T031	J13	6	937	1244	84	start	M03
164	T010	J03	0	1250	835	136	finish	M03
171	T007	J03	0	1250	835	143	finish	M02
171	T027	J09	8	1874	199	114	finish	M03
172	T017	J03	0	1250	835	144	finish	M04
175	T013	J03	0	1250	835	147	finish	M04
176	T003	J03	0	1250	835	148	finish	M01
195	T006	J03	0	1250	835	167	finish	M02
203	T008	J03	0	1250	835	175	finish	M02
210	T032	J14	1	1562	1553	97	submit	-
210	T032	J14	1	1562	1553	97	start	M02
224	T009	J03	0	1250	835	196	finish	M02
227	T011	J03	0	1250	835	199	finish	M03
228	T004	J03	0	1250	835	200	finish	M01
232	T030	J12	1	1562	1553	116	finish	M08
238	T031	J13	6	937	1244	84	finish	M03
245	T014	J03	0	1250	835	217	finish	M04
246	T005	J03	0	1250	835	218	finish	M02
248	T033	J15	1	1250	1553	52	submit	-
248	T033	J15	1	1250	1553	52	start	M01
250	T034	J16	0	905	227	254	submit	-
250	T035	J17	0	1250	197	50	submit	-
250	T034	J16	0	905	227	254	start	M01
250	T035	J17	0	1250	197	50	start	M01
252	T036	J18	1	1250	1553	254	submit	-
252	T036	J18	1	1250	1553	254	start	M02
262	T016	J03	0	1250	835	234	finish	M04
284	T037	J19	0	6250	1244	123	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
284	T037	J19	0	6250	1244	123	start	M03
297	T024	J06	4	625	311	240	finish	M01
297	T038	J20	1	1562	1553	227	submit	-
297	T038	J20	1	1562	1553	227	start	M02
300	T033	J15	1	1250	1553	52	finish	M01
300	T035	J17	0	1250	197	50	finish	M01
300	T039	J21	4	625	155	24	submit	-
300	T039	J21	4	625	155	24	start	M01
307	T032	J14	1	1562	1553	97	finish	M02
318	T040	J22	8	625	1553	1276	submit	-
318	T041	J23	4	4999	699	187	submit	-
318	T040	J22	8	625	1553	1276	start	M01
318	T041	J23	4	4999	699	187	start	M04
324	T039	J21	4	625	155	24	finish	M01
334	T018	J03	0	1250	835	306	finish	M05
336	T015	J03	0	1250	835	308	finish	M04
337	T001	J01	0	736	317	311	finish	M01
344	T042	J24	1	1250	1553	169	submit	-
344	T043	J25	0	1250	16	27	submit	-
344	T042	J24	1	1250	1553	169	start	M02
344	T043	J25	0	1250	16	27	start	M01
362	T044	J26	4	1250	466	334	submit	-
362	T044	J26	4	1250	466	334	start	M01
364	T045	J27	4	722	197	314	submit	-
364	T045	J27	4	722	197	314	start	M01
371	T043	J25	0	1250	16	27	finish	M01
375	T002	J02	4	1250	466	349	finish	M01
375	T046	J28	1	1562	1553	633	submit	-
375	T046	J28	1	1562	1553	633	start	M02
380	T047	J29	4	1477	197	297	submit	-
380	T048	J29	4	1477	197	282	submit	-
380	T049	J29	4	1477	197	278	submit	-
380	T050	J29	4	1477	197	278	submit	-
380	T047	J29	4	1477	197	297	start	M01
380	T048	J29	4	1477	197	282	start	M05
380	T049	J29	4	1477	197	278	start	M05
380	T050	J29	4	1477	197	278	start	M05
385	T051	J30	9	6250	1590	86	submit	-
385	T051	J30	9	6250	1590	86	start	M08
390	T052	J31	1	1562	1553	247	submit	-
390	T052	J31	1	1562	1553	247	start	M05
398	T053	J32	0	1250	610	1134	submit	-
398	T054	J32	0	1250	610	1124	submit	-
398	T055	J32	0	1250	610	1125	submit	-
398	T056	J32	0	1250	610	1122	submit	-
398	T057	J32	0	1250	610	1122	submit	-
398	T058	J32	0	1250	610	1121	submit	-
398	T059	J32	0	1250	610	1122	submit	-
398	T060	J32	0	1250	610	1123	submit	-
398	T061	J32	0	1250	610	1122	submit	-
398	T062	J32	0	1250	610	1121	submit	-
398	T063	J32	0	1250	610	1123	submit	-
398	T064	J32	0	1250	610	81	submit	-
398	T053	J32	0	1250	610	1134	start	M04
398	T054	J32	0	1250	610	1124	start	M08
398	T055	J32	0	1250	610	1125	start	M08
398	T056	J32	0	1250	610	1122	start	M08

time	task	job	priority	CPU	memory	run time	event	machine
398	T057	J32	0	1250	610	1122	start	M08
398	T058	J32	0	1250	610	1121	start	M08
407	T037	J19	0	6250	1244	123	finish	M03
407	T059	J32	0	1250	610	1122	start	M03
407	T060	J32	0	1250	610	1123	start	M03
407	T061	J32	0	1250	610	1122	start	M03
407	T062	J32	0	1250	610	1121	start	M03
407	T063	J32	0	1250	610	1123	start	M03
415	T021	J05	4	3125	777	358	finish	M06
415	T064	J32	0	1250	610	81	start	M06
416	T023	J05	4	3125	777	359	finish	M07
420	T022	J05	4	3125	777	363	finish	M07
428	T020	J05	4	3125	777	371	finish	M06
435	T065	J33	1	1250	1553	557	submit	-
435	T065	J33	1	1250	1553	557	start	M06
437	T066	J34	1	1250	1553	182	submit	-
437	T066	J34	1	1250	1553	182	start	M06
439	T067	J35	0	1250	932	164	submit	-
439	T068	J35	0	1250	932	153	submit	-
439	T069	J35	0	1250	932	191	submit	-
439	T070	J35	0	1250	932	209	submit	-
439	T071	J35	0	1250	932	191	submit	-
439	T072	J35	0	1250	932	149	submit	-
439	T073	J35	0	1250	932	552	submit	-
439	T074	J35	0	1250	932	194	submit	-
439	T075	J35	0	1250	932	152	submit	-
439	T076	J35	0	1250	932	149	submit	-
439	T077	J35	0	1250	932	152	submit	-
439	T078	J35	0	1250	932	153	submit	-
439	T079	J35	0	1250	932	150	submit	-
439	T080	J35	0	1250	932	173	submit	-
439	T081	J35	0	1250	932	185	submit	-
439	T082	J35	0	1250	932	163	submit	-
439	T067	J35	0	1250	932	164	start	M06
439	T068	J35	0	1250	932	153	start	M06
439	T069	J35	0	1250	932	191	start	M07
439	T070	J35	0	1250	932	209	start	M07
439	T071	J35	0	1250	932	191	start	M07
449	T041	J23	4	4999	699	187	fail	M04
449	T053	J32	0	1250	610	1134	fail	M04
471	T051	J30	9	6250	1590	86	finish	M08
471	T041	J23	4	4999	699	187	start	M08
471	T053	J32	0	1250	610	1134	start	M08
484	T083	J36	6	625	280	142	submit	-
484	T083	J36	6	625	280	142	start	M01
485	T084	J37	4	2930	2704	895	submit	-
485	T085	J37	4	2930	2704	893	submit	-
485	T086	J37	4	2930	2704	908	submit	-
485	T087	J37	4	2930	2704	898	submit	-
496	T064	J32	0	1250	610	81	finish	M06
496	T072	J35	0	1250	932	149	start	M06
504	T034	J16	0	905	227	254	finish	M01
506	T036	J18	1	1250	1553	254	finish	M02
513	T042	J24	1	1250	1553	169	finish	M02
513	T084	J37	4	2930	2704	895	start	M02
520	T088	J38	4	1250	466	263	submit	-
520	T089	J38	4	1250	466	263	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
520	T090	J38	4	1250	466	265	submit	-
520	T091	J38	4	1250	466	268	submit	-
524	T038	J20	1	1562	1553	227	finish	M02
524	T088	J38	4	1250	466	263	start	M02
525	T092	J39	9	4688	6360	580	submit	-
525	T053	J32	0	1250	610	1134	evict	M08
525	T058	J32	0	1250	610	1121	evict	M08
525	T057	J32	0	1250	610	1122	evict	M08
525	T056	J32	0	1250	610	1122	evict	M08
525	T092	J39	9	4688	6360	580	start	M08
535	T093	J40	1	1562	1553	121	submit	-
536	T094	J41	6	937	199	71	submit	-
536	T095	J41	6	937	199	74	submit	-
536	T096	J41	6	937	199	70	submit	-
536	T097	J41	6	937	199	71	submit	-
536	T098	J42	8	1250	932	357	submit	-
536	T094	J41	6	937	199	71	start	M01
536	T095	J41	6	937	199	74	start	M07
554	T099	J43	1	1562	1553	249	submit	-
582	T100	J44	1	1250	1553	205	submit	-
592	T068	J35	0	1250	932	153	finish	M06
592	T098	J42	8	1250	932	357	start	M06
596	T101	J45	8	625	199	627	submit	-
603	T067	J35	0	1250	932	164	finish	M06
603	T101	J45	8	625	199	627	start	M06
607	T094	J41	6	937	199	71	finish	M01
607	T096	J41	6	937	199	70	start	M01
610	T095	J41	6	937	199	74	finish	M07
610	T097	J41	6	937	199	71	start	M07
619	T066	J34	1	1250	1553	182	finish	M06
626	T083	J36	6	625	280	142	finish	M01
630	T069	J35	0	1250	932	191	finish	M07
630	T071	J35	0	1250	932	191	finish	M07
630	T089	J38	4	1250	466	263	start	M07
630	T090	J38	4	1250	466	265	start	M07
630	T091	J38	4	1250	466	268	start	M07
637	T052	J31	1	1562	1553	247	finish	M05
637	T093	J40	1	1562	1553	121	start	M05
645	T072	J35	0	1250	932	149	finish	M06
645	T102	J46	9	6250	3180	210	submit	-
645	T085	J37	4	2930	2704	893	start	M06
648	T070	J35	0	1250	932	209	finish	M07
648	T053	J32	0	1250	610	1134	start	M07
658	T041	J23	4	4999	699	187	finish	M08
658	T049	J29	4	1477	197	278	finish	M05
658	T050	J29	4	1477	197	278	finish	M05
658	T055	J32	0	1250	610	1125	evict	M08
658	T102	J46	9	6250	3180	210	start	M08
662	T048	J29	4	1477	197	282	finish	M05
677	T047	J29	4	1477	197	297	finish	M01
677	T096	J41	6	937	199	70	finish	M01
677	T055	J32	0	1250	610	1125	start	M01
678	T045	J27	4	722	197	314	finish	M01
681	T097	J41	6	937	199	71	finish	M07
681	T056	J32	0	1250	610	1122	start	M07
696	T044	J26	4	1250	466	334	finish	M01
696	T057	J32	0	1250	610	1122	start	M01

time	task	job	priority	CPU	memory	run time	event	machine
758	T093	J40	1	1562	1553	121	finish	M05
758	T086	J37	4	2930	2704	908	start	M05
776	T103	J47	4	1250	466	318	submit	-
777	T104	J48	4	1250	466	441	submit	-
777	T105	J49	4	1250	466	321	submit	-
780	T106	J50	4	1250	466	377	submit	-
780	T107	J50	4	1250	466	387	submit	-
780	T108	J50	4	1250	466	376	submit	-
780	T109	J50	4	1250	466	410	submit	-
787	T088	J38	4	1250	466	263	finish	M02
787	T103	J47	4	1250	466	318	start	M02
828	T110	J51	1	6250	6360	802	submit	-
829	T111	J52	1	6250	6360	784	submit	-
868	T102	J46	9	6250	3180	210	finish	M08
868	T087	J37	4	2930	2704	898	start	M08
868	T104	J48	4	1250	466	441	start	M08
868	T105	J49	4	1250	466	321	start	M08
893	T089	J38	4	1250	466	263	finish	M07
893	T106	J50	4	1250	466	377	start	M07
895	T090	J38	4	1250	466	265	finish	M07
895	T107	J50	4	1250	466	387	start	M07
898	T091	J38	4	1250	466	268	finish	M07
898	T112	J53	4	3125	1125	101	submit	-
898	T108	J50	4	1250	466	376	start	M07
945	T113	J54	4	1250	466	283	submit	-
945	T114	J55	1	5627	199	48	submit	-
949	T098	J42	8	1250	932	357	finish	M06
949	T109	J50	4	1250	466	410	start	M06
964	T115	J56	4	3125	1125	47	submit	-
964	T116	J57	4	3125	1125	77	submit	-
964	T117	J58	4	1250	398	43	submit	-
964	T117	J58	4	1250	398	43	start	M05
992	T065	J33	1	1250	1553	557	finish	M06
992	T113	J54	4	1250	466	283	start	M06
1007	T117	J58	4	1250	398	43	finish	M05
1008	T046	J28	1	1562	1553	633	finish	M02
1067	T118	J59	1	625	233	343	submit	-
1067	T118	J59	1	625	233	343	start	M01
1105	T092	J39	9	4688	6360	580	finish	M08
1105	T103	J47	4	1250	466	318	finish	M02
1105	T112	J53	4	3125	1125	101	start	M02
1105	T115	J56	4	3125	1125	47	start	M08
1105	T099	J43	1	1562	1553	249	start	M08
1106	T119	J60	9	6250	1590	230	submit	-
1106	T063	J32	0	1250	610	1123	evict	M03
1106	T062	J32	0	1250	610	1121	evict	M03
1106	T061	J32	0	1250	610	1122	evict	M03
1106	T060	J32	0	1250	610	1123	evict	M03
1106	T059	J32	0	1250	610	1122	evict	M03
1106	T119	J60	9	6250	1590	230	start	M03
1117	T120	J61	1	769	466	279	submit	-
1117	T120	J61	1	769	466	279	start	M08
1119	T121	J62	0	625	155	274	submit	-
1119	T122	J62	0	625	155	269	submit	-
1119	T121	J62	0	625	155	274	start	M05
1119	T122	J62	0	625	155	269	start	M05
1121	T123	J63	0	625	155	267	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1121	T124	J63	0	625	155	272	submit	-
1121	T125	J63	0	625	155	268	submit	-
1121	T126	J63	0	625	155	269	submit	-
1121	T127	J63	0	625	155	266	submit	-
1121	T128	J63	0	625	155	271	submit	-
1121	T129	J63	0	625	155	272	submit	-
1121	T130	J63	0	625	155	266	submit	-
1121	T131	J63	0	625	155	267	submit	-
1121	T132	J63	0	625	155	269	submit	-
1121	T133	J63	0	625	155	269	submit	-
1121	T134	J63	0	625	155	268	submit	-
1121	T135	J63	0	625	155	272	submit	-
1121	T136	J63	0	625	155	268	submit	-
1121	T137	J63	0	625	155	266	submit	-
1121	T138	J63	0	625	155	272	submit	-
1121	T139	J64	0	625	155	247	submit	-
1121	T140	J64	0	625	155	246	submit	-
1121	T141	J64	0	625	155	248	submit	-
1121	T142	J64	0	625	155	260	submit	-
1121	T143	J64	0	625	155	244	submit	-
1121	T144	J64	0	625	155	265	submit	-
1121	T145	J64	0	625	155	259	submit	-
1121	T146	J64	0	625	155	245	submit	-
1121	T147	J64	0	625	155	268	submit	-
1121	T148	J64	0	625	155	255	submit	-
1121	T149	J64	0	625	155	245	submit	-
1121	T150	J64	0	625	155	245	submit	-
1121	T151	J64	0	625	155	265	submit	-
1121	T152	J64	0	625	155	256	submit	-
1121	T153	J64	0	625	155	243	submit	-
1121	T154	J64	0	625	155	258	submit	-
1128	T155	J65	1	6250	6360	2453	submit	-
1152	T115	J56	4	3125	1125	47	finish	M08
1152	T116	J57	4	3125	1125	77	start	M08
1186	T156	J66	1	625	311	39	submit	-
1189	T105	J49	4	1250	466	321	finish	M08
1189	T100	J44	1	1250	1553	205	start	M08

Cell window 12: Events

Time 449: machine M04 failed; tasks T041, T053 stopped and went back to the queue.

Time 471: task T041 restarted on machine M08.

Time 471: task T053 restarted on machine M08.

Time 525: task T092 (priority 9) evicted task T053 (priority 0) from machine M08.

Time 525: task T092 (priority 9) evicted task T056 (priority 0) from machine M08.

Time 525: task T092 (priority 9) evicted task T057 (priority 0) from machine M08.

Time 525: task T092 (priority 9) evicted task T058 (priority 0) from machine M08.

Time 648: task T053 restarted on machine M07.

Time 658: task T102 (priority 9) evicted task T055 (priority 0) from machine M08.

Time 677: task T055 restarted on machine M01.

Time 681: task T056 restarted on machine M07.

Time 696: task T057 restarted on machine M01.

Time 1106: task T119 (priority 9) evicted task T059 (priority 0) from machine M03.

Time 1106: task T119 (priority 9) evicted task T060 (priority 0) from machine M03.

Time 1106: task T119 (priority 9) evicted task T061 (priority 0) from machine M03.

Time 1106: task T119 (priority 9) evicted task T062 (priority 0) from machine M03.

Time 1106: task T119 (priority 9) evicted task T063 (priority 0) from machine M03.

Longest waits (submission to first start, tasks started by the window end): task T100 waited 607 s (submitted at 582, first started at 1189); task T099 waited 551 s (submitted at 554, first started at 1105); task T087 waited 383 s (submitted at 485, first started at 868); task T086 waited 273 s (submitted at 485, first started at 758); task T112 waited 207 s (submitted at 898, first started at 1105).

Cell window 12: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T040	J22	8	625	1553	M01	318	394
T055	J32	0	1250	610	M01	677	602
T057	J32	0	1250	610	M01	696	618
T118	J59	1	625	233	M01	1067	210
T084	J37	4	2930	2704	M02	513	208
T112	J53	4	3125	1125	M02	1105	6
T119	J60	9	6250	1590	M03	1106	136
T086	J37	4	2930	2704	M05	758	466
T121	J62	0	625	155	M05	1119	193
T122	J62	0	625	155	M05	1119	188
T085	J37	4	2930	2704	M06	645	338
T101	J45	8	625	199	M06	603	30
T109	J50	4	1250	466	M06	949	159
T113	J54	4	1250	466	M06	992	75
T053	J32	0	1250	610	M07	648	582
T056	J32	0	1250	610	M07	681	603
T106	J50	4	1250	466	M07	893	70
T107	J50	4	1250	466	M07	895	82
T108	J50	4	1250	466	M07	898	74
T054	J32	0	1250	610	M08	398	322
T087	J37	4	2930	2704	M08	868	566
T099	J43	1	1562	1553	M08	1105	154
T100	J44	1	1250	1553	M08	1189	194
T104	J48	4	1250	466	M08	868	109
T116	J57	4	3125	1125	M08	1152	29
T120	J61	1	769	466	M08	1117	196

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T110	J51	1	6250	6360	828	802
T111	J52	1	6250	6360	829	784
T114	J55	1	5627	199	945	48
T155	J65	1	6250	6360	1128	2453
T156	J66	1	625	311	1186	39
T058	J32	0	1250	610	398	1121
T059	J32	0	1250	610	398	1122
T060	J32	0	1250	610	398	1123

task	job	priority	CPU	memory	submitted	run time
T061	J32	0	1250	610	398	1122
T062	J32	0	1250	610	398	1121
T063	J32	0	1250	610	398	1123
T073	J35	0	1250	932	439	552
T074	J35	0	1250	932	439	194
T075	J35	0	1250	932	439	152
T076	J35	0	1250	932	439	149
T077	J35	0	1250	932	439	152
T078	J35	0	1250	932	439	153
T079	J35	0	1250	932	439	150
T080	J35	0	1250	932	439	173
T081	J35	0	1250	932	439	185
T082	J35	0	1250	932	439	163
T123	J63	0	625	155	1121	267
T124	J63	0	625	155	1121	272
T125	J63	0	625	155	1121	268
T126	J63	0	625	155	1121	269
T127	J63	0	625	155	1121	266
T128	J63	0	625	155	1121	271
T129	J63	0	625	155	1121	272
T130	J63	0	625	155	1121	266
T131	J63	0	625	155	1121	267
T132	J63	0	625	155	1121	269
T133	J63	0	625	155	1121	269
T134	J63	0	625	155	1121	268
T135	J63	0	625	155	1121	272
T136	J63	0	625	155	1121	268
T137	J63	0	625	155	1121	266
T138	J63	0	625	155	1121	272
T139	J64	0	625	155	1121	247
T140	J64	0	625	155	1121	246
T141	J64	0	625	155	1121	248
T142	J64	0	625	155	1121	260
T143	J64	0	625	155	1121	244
T144	J64	0	625	155	1121	265
T145	J64	0	625	155	1121	259
T146	J64	0	625	155	1121	245
T147	J64	0	625	155	1121	268
T148	J64	0	625	155	1121	255
T149	J64	0	625	155	1121	245
T150	J64	0	625	155	1121	245
T151	J64	0	625	155	1121	265
T152	J64	0	625	155	1121	256
T153	J64	0	625	155	1121	243
T154	J64	0	625	155	1121	258

Machine M04 is down at time 1200; it returns at time 1453.

Cell window 13

Times 0 to 1200. The cluster has 8 machines. 111 tasks of 55 jobs are submitted during the window.

Cell window 13: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	3116	-	-
M03	6250	6244	-	-
M04	6250	3116	555	1553
M05	12500	12500	-	-
M06	6250	6244	-	-
M07	6250	6244	-	-
M08	6250	6244	-	-

Cell window 13: Task log

time	task	job	priority	CPU	memory	run time	event	machine
38	T001	J01	8	1874	777	581	submit	-
38	T001	J01	8	1874	777	581	start	M01
53	T002	J02	1	1250	1553	50	submit	-
53	T002	J02	1	1250	1553	50	start	M01
82	T003	J03	8	937	466	78	submit	-
82	T003	J03	8	937	466	78	start	M01
84	T004	J04	4	625	217	67	submit	-
84	T005	J04	4	625	217	67	submit	-
84	T006	J04	4	625	217	72	submit	-
84	T007	J04	4	625	217	67	submit	-
84	T004	J04	4	625	217	67	start	M01
84	T005	J04	4	625	217	67	start	M02
84	T006	J04	4	625	217	72	start	M02
84	T007	J04	4	625	217	67	start	M02
89	T008	J05	1	1562	1553	365	submit	-
89	T008	J05	1	1562	1553	365	start	M02
95	T009	J06	8	625	155	88	submit	-
95	T009	J06	8	625	155	88	start	M02
103	T002	J02	1	1250	1553	50	finish	M01
124	T010	J07	4	1250	466	349	submit	-
124	T010	J07	4	1250	466	349	start	M01
129	T011	J08	1	1562	1553	105	submit	-
129	T011	J08	1	1562	1553	105	start	M03
145	T012	J09	0	6250	466	44	submit	-
145	T012	J09	0	6250	466	44	start	M04
149	T013	J10	1	1250	1590	432	submit	-
149	T013	J10	1	1250	1590	432	start	M03
151	T004	J04	4	625	217	67	finish	M01
151	T005	J04	4	625	217	67	finish	M02
151	T007	J04	4	625	217	67	finish	M02
153	T014	J11	1	3125	2386	245	submit	-
153	T014	J11	1	3125	2386	245	start	M03
155	T015	J12	4	3125	777	371	submit	-
155	T016	J12	4	3125	777	358	submit	-
155	T017	J12	4	3125	777	363	submit	-
155	T018	J12	4	3125	777	359	submit	-
155	T015	J12	4	3125	777	371	start	M02
155	T016	J12	4	3125	777	358	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
155	T017	J12	4	3125	777	363	start	M05
155	T018	J12	4	3125	777	359	start	M05
156	T006	J04	4	625	217	72	finish	M02
160	T003	J03	8	937	466	78	finish	M01
173	T019	J13	8	6250	1590	1278	submit	-
173	T019	J13	8	6250	1590	1278	start	M06
183	T009	J06	8	625	155	88	finish	M02
189	T012	J09	0	6250	466	44	finish	M04
214	T020	J14	1	1562	1553	116	submit	-
214	T020	J14	1	1562	1553	116	start	M01
215	T021	J15	4	3125	1125	48	submit	-
215	T021	J15	4	3125	1125	48	start	M04
234	T011	J08	1	1562	1553	105	finish	M03
234	T022	J16	4	937	1035	113	submit	-
234	T023	J16	4	937	1035	82	submit	-
234	T024	J16	4	937	1035	81	submit	-
234	T025	J16	4	937	1035	98	submit	-
234	T026	J17	8	1250	777	61	submit	-
234	T026	J17	8	1250	777	61	start	M02
234	T022	J16	4	937	1035	113	start	M03
234	T023	J16	4	937	1035	82	start	M03
234	T024	J16	4	937	1035	81	start	M04
234	T025	J16	4	937	1035	98	start	M05
263	T021	J15	4	3125	1125	48	finish	M04
279	T027	J18	1	1250	1553	37	submit	-
279	T027	J18	1	1250	1553	37	start	M04
294	T028	J19	1	1562	1553	325	submit	-
294	T028	J19	1	1562	1553	325	start	M05
295	T026	J17	8	1250	777	61	finish	M02
308	T029	J20	1	1562	1553	97	submit	-
308	T029	J20	1	1562	1553	97	start	M07
315	T024	J16	4	937	1035	81	finish	M04
316	T023	J16	4	937	1035	82	finish	M03
316	T027	J18	1	1250	1553	37	finish	M04
330	T020	J14	1	1562	1553	116	finish	M01
332	T025	J16	4	937	1035	98	finish	M05
347	T022	J16	4	937	1035	113	finish	M03
348	T030	J21	0	905	227	254	submit	-
348	T030	J21	0	905	227	254	start	M01
352	T031	J22	0	1250	283	252	submit	-
352	T031	J22	0	1250	283	252	start	M01
354	T032	J23	0	1250	217	147	submit	-
354	T033	J23	0	1250	217	223	submit	-
354	T034	J23	0	1250	217	188	submit	-
354	T035	J23	0	1250	217	168	submit	-
354	T036	J23	0	1250	217	205	submit	-
354	T037	J23	0	1250	217	246	submit	-
354	T038	J23	0	1250	217	211	submit	-
354	T039	J23	0	1250	217	183	submit	-
354	T040	J23	0	1250	217	178	submit	-
354	T041	J23	0	1250	217	164	submit	-
354	T042	J23	0	1250	217	201	submit	-
354	T043	J23	0	1250	217	247	submit	-
354	T044	J23	0	1250	217	231	submit	-
354	T045	J23	0	1250	217	248	submit	-
354	T032	J23	0	1250	217	147	start	M02
354	T033	J23	0	1250	217	223	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
354	T034	J23	0	1250	217	188	start	M04
354	T035	J23	0	1250	217	168	start	M04
354	T036	J23	0	1250	217	205	start	M04
354	T037	J23	0	1250	217	246	start	M04
354	T038	J23	0	1250	217	211	start	M04
354	T039	J23	0	1250	217	183	start	M05
354	T040	J23	0	1250	217	178	start	M07
354	T041	J23	0	1250	217	164	start	M07
354	T042	J23	0	1250	217	201	start	M07
354	T043	J23	0	1250	217	247	start	M08
354	T044	J23	0	1250	217	231	start	M08
354	T045	J23	0	1250	217	248	start	M08
382	T046	J24	0	6250	1244	123	submit	-
398	T014	J11	1	3125	2386	245	finish	M03
398	T047	J25	4	625	155	24	submit	-
398	T047	J25	4	625	155	24	start	M01
405	T029	J20	1	1562	1553	97	finish	M07
422	T047	J25	4	625	155	24	finish	M01
445	T048	J26	0	1250	197	152	submit	-
445	T049	J26	0	1250	197	106	submit	-
445	T050	J26	0	1250	197	127	submit	-
445	T048	J26	0	1250	197	152	start	M03
445	T049	J26	0	1250	197	106	start	M03
445	T050	J26	0	1250	197	127	start	M03
454	T008	J05	1	1562	1553	365	finish	M02
473	T010	J07	4	1250	466	349	finish	M01
483	T051	J27	9	6250	1590	86	submit	-
483	T050	J26	0	1250	197	127	evict	M03
483	T049	J26	0	1250	197	106	evict	M03
483	T048	J26	0	1250	197	152	evict	M03
483	T033	J23	0	1250	217	223	evict	M03
483	T013	J10	1	1250	1590	432	evict	M03
483	T051	J27	9	6250	1590	86	start	M03
483	T013	J10	1	1250	1590	432	start	M01
483	T033	J23	0	1250	217	223	start	M02
483	T048	J26	0	1250	197	152	start	M07
483	T049	J26	0	1250	197	106	start	M07
483	T050	J26	0	1250	197	127	start	M08
488	T052	J28	1	1562	1553	247	submit	-
493	T053	J29	1	937	466	1137	submit	-
493	T053	J29	1	937	466	1137	start	M08
496	T054	J30	0	1250	610	1134	submit	-
496	T055	J30	0	1250	610	1124	submit	-
496	T056	J30	0	1250	610	1125	submit	-
496	T057	J30	0	1250	610	1122	submit	-
496	T058	J30	0	1250	610	1122	submit	-
496	T059	J30	0	1250	610	1121	submit	-
496	T060	J30	0	1250	610	1122	submit	-
496	T061	J30	0	1250	610	1123	submit	-
496	T062	J30	0	1250	610	1122	submit	-
496	T063	J30	0	1250	610	1121	submit	-
496	T064	J30	0	1250	610	1123	submit	-
496	T065	J30	0	1250	610	81	submit	-
501	T032	J23	0	1250	217	147	finish	M02
501	T052	J28	1	1562	1553	247	start	M02
513	T016	J12	4	3125	777	358	finish	M05
514	T018	J12	4	3125	777	359	finish	M05

time	task	job	priority	CPU	memory	run time	event	machine
514	T046	J24	0	6250	1244	123	start	M05
516	T066	J31	8	625	155	165	submit	-
516	T066	J31	8	625	155	165	start	M01
518	T017	J12	4	3125	777	363	finish	M05
518	T041	J23	0	1250	217	164	finish	M07
518	T054	J30	0	1250	610	1134	start	M05
518	T055	J30	0	1250	610	1124	start	M05
518	T056	J30	0	1250	610	1125	start	M07
522	T035	J23	0	1250	217	168	finish	M04
522	T057	J30	0	1250	610	1122	start	M04
526	T015	J12	4	3125	777	371	finish	M02
526	T058	J30	0	1250	610	1122	start	M02
526	T059	J30	0	1250	610	1121	start	M02
532	T040	J23	0	1250	217	178	finish	M07
532	T060	J30	0	1250	610	1122	start	M07
533	T067	J32	1	1250	1553	557	submit	-
535	T068	J33	1	1250	1553	182	submit	-
537	T039	J23	0	1250	217	183	finish	M05
537	T067	J32	1	1250	1553	557	start	M05
542	T034	J23	0	1250	217	188	finish	M04
542	T068	J33	1	1250	1553	182	start	M04
555	T042	J23	0	1250	217	201	finish	M07
555	T036	J23	0	1250	217	205	fail	M04
555	T037	J23	0	1250	217	246	fail	M04
555	T038	J23	0	1250	217	211	fail	M04
555	T057	J30	0	1250	610	1122	fail	M04
555	T068	J33	1	1250	1553	182	fail	M04
555	T068	J33	1	1250	1553	182	start	M07
569	T051	J27	9	6250	1590	86	finish	M03
569	T036	J23	0	1250	217	205	start	M03
569	T037	J23	0	1250	217	246	start	M03
569	T038	J23	0	1250	217	211	start	M03
569	T057	J30	0	1250	610	1122	start	M03
569	T061	J30	0	1250	610	1123	start	M03
583	T069	J34	4	2487	2704	1088	submit	-
583	T070	J34	4	2487	2704	1119	submit	-
583	T071	J34	4	2487	2704	1119	submit	-
583	T072	J34	4	2487	2704	1084	submit	-
585	T044	J23	0	1250	217	231	finish	M08
589	T049	J26	0	1250	197	106	finish	M07
589	T062	J30	0	1250	610	1122	start	M07
601	T043	J23	0	1250	217	247	finish	M08
601	T069	J34	4	2487	2704	1088	start	M08
602	T030	J21	0	905	227	254	finish	M01
602	T045	J23	0	1250	217	248	finish	M08
604	T031	J22	0	1250	283	252	finish	M01
610	T050	J26	0	1250	197	127	finish	M08
610	T070	J34	4	2487	2704	1119	start	M08
616	T073	J35	6	937	1244	92	submit	-
616	T073	J35	6	937	1244	92	start	M05
619	T001	J01	8	1874	777	581	finish	M01
619	T028	J19	1	1562	1553	325	finish	M05
619	T063	J30	0	1250	610	1121	start	M01
619	T064	J30	0	1250	610	1123	start	M01
619	T065	J30	0	1250	610	81	start	M05
623	T074	J36	9	4688	6360	580	submit	-
623	T065	J30	0	1250	610	81	evict	M05

time	task	job	priority	CPU	memory	run time	event	machine
623	T055	J30	0	1250	610	1124	evict	M05
623	T054	J30	0	1250	610	1134	evict	M05
623	T046	J24	0	6250	1244	123	evict	M05
623	T074	J36	9	4688	6360	580	start	M05
623	T071	J34	4	2487	2704	1119	start	M05
623	T054	J30	0	1250	610	1134	start	M05
635	T048	J26	0	1250	197	152	finish	M07
635	T065	J30	0	1250	610	81	start	M07
652	T075	J37	1	1562	1553	249	submit	-
680	T076	J38	1	1250	1553	205	submit	-
680	T077	J39	1	1250	1553	57	submit	-
681	T066	J31	8	625	155	165	finish	M01
682	T078	J40	0	1250	197	55	submit	-
682	T079	J41	0	3125	311	617	submit	-
682	T078	J40	0	1250	197	55	start	M01
706	T033	J23	0	1250	217	223	finish	M02
708	T073	J35	6	937	1244	92	finish	M05
708	T055	J30	0	1250	610	1124	start	M05
716	T065	J30	0	1250	610	81	finish	M07
737	T068	J33	1	1250	1553	182	finish	M07
737	T078	J40	0	1250	197	55	finish	M01
737	T072	J34	4	2487	2704	1084	start	M07
743	T080	J42	0	6250	466	41	submit	-
743	T081	J43	9	6250	3180	210	submit	-
743	T061	J30	0	1250	610	1123	evict	M03
743	T057	J30	0	1250	610	1122	evict	M03
743	T038	J23	0	1250	217	211	evict	M03
743	T037	J23	0	1250	217	246	evict	M03
743	T036	J23	0	1250	217	205	evict	M03
743	T081	J43	9	6250	3180	210	start	M03
743	T036	J23	0	1250	217	205	start	M01
743	T037	J23	0	1250	217	246	start	M02
743	T038	J23	0	1250	217	211	start	M05
744	T082	J44	0	12500	7959	35	submit	-
744	T083	J44	0	12500	7959	29	submit	-
744	T084	J44	0	12500	7959	36	submit	-
744	T085	J44	0	12500	7959	29	submit	-
744	T086	J44	0	12500	7959	28	submit	-
744	T087	J44	0	12500	7959	28	submit	-
744	T088	J44	0	12500	7959	20	submit	-
744	T089	J44	0	12500	7959	26	submit	-
744	T090	J44	0	12500	7959	24	submit	-
744	T091	J44	0	12500	7959	24	submit	-
744	T092	J44	0	12500	7959	29	submit	-
744	T093	J44	0	12500	7959	25	submit	-
744	T094	J44	0	12500	7959	30	submit	-
744	T095	J44	0	12500	7959	35	submit	-
744	T096	J44	0	12500	7959	20	submit	-
744	T097	J44	0	12500	7959	26	submit	-
748	T052	J28	1	1562	1553	247	finish	M02
748	T075	J37	1	1562	1553	249	start	M02
864	T098	J45	0	6250	1244	77	submit	-
878	T099	J46	4	1250	466	260	submit	-
878	T100	J46	4	1250	466	266	submit	-
878	T101	J46	4	1250	466	305	submit	-
878	T102	J46	4	1250	466	302	submit	-
915	T013	J10	1	1250	1590	432	finish	M01

time	task	job	priority	CPU	memory	run time	event	machine
915	T099	J46	4	1250	466	260	start	M01
915	T100	J46	4	1250	466	266	start	M01
925	T103	J47	1	6250	6360	1312	submit	-
926	T104	J48	1	6250	6360	802	submit	-
927	T105	J49	1	6250	6360	784	submit	-
948	T036	J23	0	1250	217	205	finish	M01
948	T101	J46	4	1250	466	305	start	M01
953	T081	J43	9	6250	3180	210	finish	M03
953	T102	J46	4	1250	466	302	start	M03
953	T076	J38	1	1250	1553	205	start	M03
953	T077	J39	1	1250	1553	57	start	M03
953	T057	J30	0	1250	610	1122	start	M03
953	T061	J30	0	1250	610	1123	start	M03
954	T038	J23	0	1250	217	211	finish	M05
989	T037	J23	0	1250	217	246	finish	M02
997	T075	J37	1	1562	1553	249	finish	M02
997	T079	J41	0	3125	311	617	start	M02
1010	T077	J39	1	1250	1553	57	finish	M03
1043	T106	J50	1	5627	199	48	submit	-
1062	T107	J51	4	3125	1125	47	submit	-
1062	T108	J52	4	3125	1125	77	submit	-
1094	T067	J32	1	1250	1553	557	finish	M05
1115	T109	J53	4	3125	1125	118	submit	-
1158	T076	J38	1	1250	1553	205	finish	M03
1164	T110	J54	1	625	233	360	submit	-
1164	T110	J54	1	625	233	360	start	M02
1165	T111	J55	1	625	233	343	submit	-
1165	T111	J55	1	625	233	343	start	M03
1175	T099	J46	4	1250	466	260	finish	M01
1181	T100	J46	4	1250	466	266	finish	M01

Cell window 13: Events

Time 483: task T051 (priority 9) evicted task T013 (priority 1) from machine M03.

Time 483: task T051 (priority 9) evicted task T033 (priority 0) from machine M03.

Time 483: task T051 (priority 9) evicted task T048 (priority 0) from machine M03.

Time 483: task T051 (priority 9) evicted task T049 (priority 0) from machine M03.

Time 483: task T051 (priority 9) evicted task T050 (priority 0) from machine M03.

Time 483: task T013 restarted on machine M01.

Time 483: task T033 restarted on machine M02.

Time 483: task T048 restarted on machine M07.

Time 483: task T049 restarted on machine M07.

Time 483: task T050 restarted on machine M08.

Time 555: machine M04 failed; tasks T036, T037, T038, T057, T068 stopped and went back to the queue.

Time 555: task T068 restarted on machine M07.

Time 569: task T036 restarted on machine M03.

Time 569: task T037 restarted on machine M03.

Time 569: task T038 restarted on machine M03.

Time 569: task T057 restarted on machine M03.

Time 623: task T074 (priority 9) evicted task T046 (priority 0) from machine M05.

Time 623: task T074 (priority 9) evicted task T054 (priority 0) from machine M05.

Time 623: task T074 (priority 9) evicted task T055 (priority 0) from machine M05.

Time 623: task T074 (priority 9) evicted task T065 (priority 0) from machine M05.

Time 623: task T054 restarted on machine M05.

Time 635: task T065 restarted on machine M07.

Time 708: task T055 restarted on machine M05.

Time 743: task T081 (priority 9) evicted task T036 (priority 0) from machine M03.

Time 743: task T081 (priority 9) evicted task T037 (priority 0) from machine M03.

Time 743: task T081 (priority 9) evicted task T038 (priority 0) from machine M03.

Time 743: task T081 (priority 9) evicted task T057 (priority 0) from machine M03.

Time 743: task T081 (priority 9) evicted task T061 (priority 0) from machine M03.

Time 743: task T036 restarted on machine M01.

Time 743: task T037 restarted on machine M02.

Time 743: task T038 restarted on machine M05.

Time 953: task T057 restarted on machine M03.

Time 953: task T061 restarted on machine M03.

Longest waits (submission to first start, tasks started by the window end): task T079 waited 315 s (submitted at 682, first started at 997); task T076 waited 273 s (submitted at 680, first started at 953); task T077 waited 273 s (submitted at 680, first started at 953); task T072 waited 154 s (submitted at 583, first started at 737); task T046 waited 132 s (submitted at 382, first started at 514).

Cell window 13: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T063	J30	0	1250	610	M01	619	540
T064	J30	0	1250	610	M01	619	542
T101	J46	4	1250	466	M01	948	53
T058	J30	0	1250	610	M02	526	448
T059	J30	0	1250	610	M02	526	447
T079	J41	0	3125	311	M02	997	414
T110	J54	1	625	233	M02	1164	324
T057	J30	0	1250	610	M03	953	875
T061	J30	0	1250	610	M03	953	876
T102	J46	4	1250	466	M03	953	55
T111	J55	1	625	233	M03	1165	308
T054	J30	0	1250	610	M05	623	557
T055	J30	0	1250	610	M05	708	632
T071	J34	4	2487	2704	M05	623	542
T074	J36	9	4688	6360	M05	623	3
T019	J13	8	6250	1590	M06	173	251
T056	J30	0	1250	610	M07	518	443
T060	J30	0	1250	610	M07	532	454
T062	J30	0	1250	610	M07	589	511
T072	J34	4	2487	2704	M07	737	621
T053	J29	1	937	466	M08	493	430

task	job	priority	CPU	memory	machine	started	remaining
T069	J34	4	2487	2704	M08	601	489
T070	J34	4	2487	2704	M08	610	529

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T107	J51	4	3125	1125	1062	47
T108	J52	4	3125	1125	1062	77
T109	J53	4	3125	1125	1115	118
T103	J47	1	6250	6360	925	1312
T104	J48	1	6250	6360	926	802
T105	J49	1	6250	6360	927	784
T106	J50	1	5627	199	1043	48
T046	J24	0	6250	1244	382	123
T080	J42	0	6250	466	743	41
T082	J44	0	12500	7959	744	35
T083	J44	0	12500	7959	744	29
T084	J44	0	12500	7959	744	36
T085	J44	0	12500	7959	744	29
T086	J44	0	12500	7959	744	28
T087	J44	0	12500	7959	744	28
T088	J44	0	12500	7959	744	20
T089	J44	0	12500	7959	744	26
T090	J44	0	12500	7959	744	24
T091	J44	0	12500	7959	744	24
T092	J44	0	12500	7959	744	29
T093	J44	0	12500	7959	744	25
T094	J44	0	12500	7959	744	30
T095	J44	0	12500	7959	744	35
T096	J44	0	12500	7959	744	20
T097	J44	0	12500	7959	744	26
T098	J45	0	6250	1244	864	77

Machine M04 is down at time 1200; it returns at time 1553.

Cell window 15

Times 0 to 1200. The cluster has 8 machines. 218 tasks of 61 jobs are submitted during the window.

Cell window 15: Machines

machine	CPU	memory	fails at	returns at
M01	6250	6244	-	-
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	6250	3116	385	780
M05	6250	6244	-	-
M06	6250	6244	-	-
M07	6250	6244	-	-
M08	6250	6244	-	-

Cell window 15: Task log

time	task	job	priority	CPU	memory	run time	event	machine
38	T001	J01	1	937	1590	145	submit	-
38	T001	J01	1	937	1590	145	start	M01
39	T002	J02	1	937	1590	250	submit	-
39	T002	J02	1	937	1590	250	start	M01
40	T003	J03	0	730	1132	143	submit	-
40	T003	J03	0	730	1132	143	start	M01
41	T004	J04	0	730	1132	248	submit	-
41	T004	J04	0	730	1132	248	start	M01
42	T005	J05	0	937	1125	113	submit	-
42	T006	J05	0	937	1125	83	submit	-
42	T007	J05	0	937	1125	82	submit	-
42	T008	J05	0	937	1125	72	submit	-
42	T009	J05	0	937	1125	83	submit	-
42	T010	J05	0	937	1125	86	submit	-
42	T011	J05	0	937	1125	126	submit	-
42	T012	J05	0	937	1125	86	submit	-
42	T013	J05	0	937	1125	75	submit	-
42	T014	J05	0	937	1125	81	submit	-
42	T015	J05	0	937	1125	83	submit	-
42	T016	J05	0	937	1125	95	submit	-
42	T017	J05	0	937	1125	94	submit	-
42	T018	J05	0	937	1125	133	submit	-
42	T019	J05	0	937	1125	82	submit	-
42	T020	J05	0	937	1125	82	submit	-
42	T005	J05	0	937	1125	113	start	M02
42	T006	J05	0	937	1125	83	start	M02
42	T007	J05	0	937	1125	82	start	M02
42	T008	J05	0	937	1125	72	start	M02
42	T009	J05	0	937	1125	83	start	M02
42	T010	J05	0	937	1125	86	start	M03
42	T011	J05	0	937	1125	126	start	M03
42	T012	J05	0	937	1125	86	start	M03
42	T013	J05	0	937	1125	75	start	M03
42	T014	J05	0	937	1125	81	start	M03
42	T015	J05	0	937	1125	83	start	M04
42	T016	J05	0	937	1125	95	start	M04
42	T017	J05	0	937	1125	94	start	M05
42	T018	J05	0	937	1125	133	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
42	T019	J05	0	937	1125	82	start	M05
42	T020	J05	0	937	1125	82	start	M05
43	T021	J06	0	937	1125	89	submit	-
43	T022	J06	0	937	1125	83	submit	-
43	T023	J06	0	937	1125	88	submit	-
43	T024	J06	0	937	1125	124	submit	-
43	T025	J06	0	937	1125	143	submit	-
43	T026	J06	0	937	1125	88	submit	-
43	T027	J06	0	937	1125	106	submit	-
43	T028	J06	0	937	1125	78	submit	-
43	T029	J06	0	937	1125	97	submit	-
43	T030	J06	0	937	1125	81	submit	-
43	T031	J06	0	937	1125	76	submit	-
43	T032	J06	0	937	1125	75	submit	-
43	T033	J06	0	937	1125	87	submit	-
43	T034	J06	0	937	1125	74	submit	-
43	T035	J06	0	937	1125	78	submit	-
43	T036	J06	0	937	1125	138	submit	-
43	T037	J07	8	3125	3180	144	submit	-
43	T037	J07	8	3125	3180	144	start	M06
43	T021	J06	0	937	1125	89	start	M05
43	T022	J06	0	937	1125	83	start	M06
43	T023	J06	0	937	1125	88	start	M06
43	T024	J06	0	937	1125	124	start	M07
43	T025	J06	0	937	1125	143	start	M07
43	T026	J06	0	937	1125	88	start	M07
43	T027	J06	0	937	1125	106	start	M07
43	T028	J06	0	937	1125	78	start	M07
43	T029	J06	0	937	1125	97	start	M08
43	T030	J06	0	937	1125	81	start	M08
43	T031	J06	0	937	1125	76	start	M08
43	T032	J06	0	937	1125	75	start	M08
43	T033	J06	0	937	1125	87	start	M08
96	T038	J08	0	6250	466	40	submit	-
96	T039	J09	0	6250	1244	73	submit	-
97	T040	J10	0	6250	1244	74	submit	-
98	T041	J11	1	1250	1590	394	submit	-
106	T042	J12	4	625	311	29	submit	-
106	T042	J12	4	625	311	29	start	M01
114	T008	J05	0	937	1125	72	finish	M02
114	T041	J11	1	1250	1590	394	start	M02
117	T013	J05	0	937	1125	75	finish	M03
117	T034	J06	0	937	1125	74	start	M03
118	T032	J06	0	937	1125	75	finish	M08
118	T035	J06	0	937	1125	78	start	M08
119	T031	J06	0	937	1125	76	finish	M08
119	T036	J06	0	937	1125	138	start	M08
121	T028	J06	0	937	1125	78	finish	M07
123	T014	J05	0	937	1125	81	finish	M03
124	T007	J05	0	937	1125	82	finish	M02
124	T019	J05	0	937	1125	82	finish	M05
124	T020	J05	0	937	1125	82	finish	M05
124	T030	J06	0	937	1125	81	finish	M08
125	T006	J05	0	937	1125	83	finish	M02
125	T009	J05	0	937	1125	83	finish	M02
125	T015	J05	0	937	1125	83	finish	M04
126	T022	J06	0	937	1125	83	finish	M06

time	task	job	priority	CPU	memory	run time	event	machine
127	T043	J13	1	1250	466	194	submit	-
127	T043	J13	1	1250	466	194	start	M01
128	T010	J05	0	937	1125	86	finish	M03
128	T012	J05	0	937	1125	86	finish	M03
129	T044	J14	0	625	253	153	submit	-
129	T045	J14	0	625	253	148	submit	-
129	T046	J14	0	625	253	177	submit	-
129	T047	J14	0	625	253	157	submit	-
129	T048	J14	0	625	253	157	submit	-
129	T049	J14	0	625	253	163	submit	-
129	T050	J14	0	625	253	155	submit	-
129	T051	J14	0	625	253	161	submit	-
129	T052	J14	0	625	253	142	submit	-
129	T053	J14	0	625	253	153	submit	-
129	T054	J14	0	625	253	144	submit	-
129	T055	J14	0	625	253	160	submit	-
129	T056	J14	0	625	253	147	submit	-
129	T057	J14	0	625	253	153	submit	-
129	T058	J14	0	625	253	141	submit	-
129	T059	J14	0	625	253	157	submit	-
129	T044	J14	0	625	253	153	start	M02
129	T045	J14	0	625	253	148	start	M02
129	T046	J14	0	625	253	177	start	M02
129	T047	J14	0	625	253	157	start	M02
129	T048	J14	0	625	253	157	start	M02
129	T049	J14	0	625	253	163	start	M02
129	T050	J14	0	625	253	155	start	M03
129	T051	J14	0	625	253	161	start	M03
129	T052	J14	0	625	253	142	start	M03
129	T053	J14	0	625	253	153	start	M03
129	T054	J14	0	625	253	144	start	M03
129	T055	J14	0	625	253	160	start	M03
129	T056	J14	0	625	253	147	start	M03
129	T057	J14	0	625	253	153	start	M05
129	T058	J14	0	625	253	141	start	M05
129	T059	J14	0	625	253	157	start	M05
130	T033	J06	0	937	1125	87	finish	M08
131	T023	J06	0	937	1125	88	finish	M06
131	T026	J06	0	937	1125	88	finish	M07
132	T021	J06	0	937	1125	89	finish	M05
135	T042	J12	4	625	311	29	finish	M01
136	T017	J05	0	937	1125	94	finish	M05
137	T016	J05	0	937	1125	95	finish	M04
137	T038	J08	0	6250	466	40	start	M04
140	T029	J06	0	937	1125	97	finish	M08
145	T060	J15	4	1250	466	200	submit	-
145	T061	J15	4	1250	466	200	submit	-
145	T062	J15	4	1250	466	200	submit	-
145	T063	J15	4	1250	466	200	submit	-
145	T060	J15	4	1250	466	200	start	M05
145	T061	J15	4	1250	466	200	start	M05
145	T062	J15	4	1250	466	200	start	M06
145	T063	J15	4	1250	466	200	start	M06
149	T027	J06	0	937	1125	106	finish	M07
155	T005	J05	0	937	1125	113	finish	M02
167	T024	J06	0	937	1125	124	finish	M07
168	T011	J05	0	937	1125	126	finish	M03

time	task	job	priority	CPU	memory	run time	event	machine
175	T018	J05	0	937	1125	133	finish	M05
177	T038	J08	0	6250	466	40	finish	M04
177	T039	J09	0	6250	1244	73	start	M04
183	T001	J01	1	937	1590	145	finish	M01
183	T003	J03	0	730	1132	143	finish	M01
186	T025	J06	0	937	1125	143	finish	M07
186	T040	J10	0	6250	1244	74	start	M07
187	T037	J07	8	3125	3180	144	finish	M06
187	T064	J16	9	6250	1590	76	submit	-
187	T004	J04	0	730	1132	248	evict	M01
187	T043	J13	1	1250	466	194	evict	M01
187	T002	J02	1	937	1590	250	evict	M01
187	T064	J16	9	6250	1590	76	start	M01
187	T002	J02	1	937	1590	250	start	M02
187	T043	J13	1	1250	466	194	start	M05
187	T004	J04	0	730	1132	248	start	M03
191	T034	J06	0	937	1125	74	finish	M03
196	T035	J06	0	937	1125	78	finish	M08
238	T065	J17	9	312	522	628	submit	-
238	T065	J17	9	312	522	628	start	M02
250	T039	J09	0	6250	1244	73	finish	M04
257	T036	J06	0	937	1125	138	finish	M08
260	T040	J10	0	6250	1244	74	finish	M07
263	T064	J16	9	6250	1590	76	finish	M01
270	T058	J14	0	625	253	141	finish	M05
271	T052	J14	0	625	253	142	finish	M03
273	T054	J14	0	625	253	144	finish	M03
276	T056	J14	0	625	253	147	finish	M03
277	T045	J14	0	625	253	148	finish	M02
281	T066	J18	4	3125	777	53	submit	-
281	T066	J18	4	3125	777	53	start	M01
282	T044	J14	0	625	253	153	finish	M02
282	T053	J14	0	625	253	153	finish	M03
282	T057	J14	0	625	253	153	finish	M05
284	T050	J14	0	625	253	155	finish	M03
286	T047	J14	0	625	253	157	finish	M02
286	T048	J14	0	625	253	157	finish	M02
286	T059	J14	0	625	253	157	finish	M05
289	T055	J14	0	625	253	160	finish	M03
290	T051	J14	0	625	253	161	finish	M03
292	T049	J14	0	625	253	163	finish	M02
306	T046	J14	0	625	253	177	finish	M02
332	T067	J19	0	3125	388	274	submit	-
332	T067	J19	0	3125	388	274	start	M01
334	T066	J18	4	3125	777	53	finish	M01
345	T060	J15	4	1250	466	200	finish	M05
345	T061	J15	4	1250	466	200	finish	M05
345	T062	J15	4	1250	466	200	finish	M06
345	T063	J15	4	1250	466	200	finish	M06
373	T068	J20	4	1250	870	834	submit	-
373	T069	J20	4	1250	870	832	submit	-
373	T070	J20	4	1250	870	833	submit	-
373	T071	J20	4	1250	870	847	submit	-
373	T068	J20	4	1250	870	834	start	M01
373	T069	J20	4	1250	870	832	start	M01
373	T070	J20	4	1250	870	833	start	M02
373	T071	J20	4	1250	870	847	start	M02

time	task	job	priority	CPU	memory	run time	event	machine
381	T043	J13	1	1250	466	194	finish	M05
387	T072	J21	8	1250	1166	120	submit	-
387	T072	J21	8	1250	1166	120	start	M03
392	T073	J22	0	1250	16	19	submit	-
392	T073	J22	0	1250	16	19	start	M02
394	T074	J23	9	6250	3180	221	submit	-
394	T074	J23	9	6250	3180	221	start	M05
404	T075	J24	4	3125	448	64	submit	-
404	T075	J24	4	3125	448	64	start	M03
411	T073	J22	0	1250	16	19	finish	M02
435	T004	J04	0	730	1132	248	finish	M03
437	T002	J02	1	937	1590	250	finish	M02
443	T076	J25	8	625	398	621	submit	-
443	T076	J25	8	625	398	621	start	M01
468	T075	J24	4	3125	448	64	finish	M03
507	T072	J21	8	1250	1166	120	finish	M03
508	T041	J11	1	1250	1590	394	finish	M02
566	T077	J26	1	1562	1553	900	submit	-
566	T077	J26	1	1562	1553	900	start	M02
572	T078	J27	0	6250	1244	54	submit	-
572	T078	J27	0	6250	1244	54	start	M03
573	T079	J28	0	6250	1244	47	submit	-
573	T079	J28	0	6250	1244	47	start	M06
582	T080	J29	4	3125	1125	47	submit	-
582	T080	J29	4	3125	1125	47	start	M07
590	T081	J30	0	1250	777	1251	submit	-
590	T082	J30	0	1250	777	1098	submit	-
590	T081	J30	0	1250	777	1251	start	M02
590	T082	J30	0	1250	777	1098	start	M07
592	T083	J31	0	937	266	253	submit	-
592	T084	J31	0	937	266	109	submit	-
592	T085	J31	0	937	266	94	submit	-
592	T086	J31	0	937	266	105	submit	-
592	T087	J31	0	937	266	94	submit	-
592	T088	J31	0	937	266	104	submit	-
592	T089	J31	0	937	266	94	submit	-
592	T090	J31	0	937	266	94	submit	-
592	T091	J31	0	937	266	107	submit	-
592	T092	J31	0	937	266	92	submit	-
592	T093	J31	0	937	266	136	submit	-
592	T094	J31	0	937	266	203	submit	-
592	T095	J31	0	937	266	106	submit	-
592	T096	J31	0	937	266	117	submit	-
592	T097	J31	0	937	266	92	submit	-
592	T098	J31	0	937	266	106	submit	-
592	T083	J31	0	937	266	253	start	M07
592	T084	J31	0	937	266	109	start	M07
592	T085	J31	0	937	266	94	start	M08
592	T086	J31	0	937	266	105	start	M08
592	T087	J31	0	937	266	94	start	M08
592	T088	J31	0	937	266	104	start	M08
592	T089	J31	0	937	266	94	start	M08
592	T090	J31	0	937	266	94	start	M08
595	T099	J32	8	6250	3821	1104	submit	-
600	T100	J33	0	1250	384	853	submit	-
602	T101	J34	0	1250	1088	750	submit	-
602	T102	J34	0	1250	1088	571	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
602	T103	J34	0	1250	1088	601	submit	-
602	T104	J34	0	1250	1088	708	submit	-
602	T105	J34	0	1250	1088	582	submit	-
602	T106	J34	0	1250	1088	643	submit	-
602	T107	J34	0	1250	1088	595	submit	-
602	T108	J34	0	1250	1088	568	submit	-
602	T109	J34	0	1250	1088	636	submit	-
602	T110	J34	0	1250	1088	590	submit	-
602	T111	J34	0	1250	1088	849	submit	-
602	T112	J34	0	1250	1088	601	submit	-
602	T113	J34	0	1250	1088	761	submit	-
602	T114	J34	0	1250	1088	619	submit	-
602	T115	J34	0	1250	1088	601	submit	-
602	T116	J34	0	1250	1088	571	submit	-
606	T067	J19	0	3125	388	274	finish	M01
606	T091	J31	0	937	266	107	start	M01
606	T092	J31	0	937	266	92	start	M01
606	T093	J31	0	937	266	136	start	M01
615	T074	J23	9	6250	3180	221	finish	M05
615	T099	J32	8	6250	3821	1104	start	M05
620	T079	J28	0	6250	1244	47	finish	M06
620	T094	J31	0	937	266	203	start	M06
620	T095	J31	0	937	266	106	start	M06
620	T096	J31	0	937	266	117	start	M06
620	T097	J31	0	937	266	92	start	M06
620	T098	J31	0	937	266	106	start	M06
620	T100	J33	0	1250	384	853	start	M06
626	T078	J27	0	6250	1244	54	finish	M03
626	T101	J34	0	1250	1088	750	start	M03
626	T102	J34	0	1250	1088	571	start	M03
626	T103	J34	0	1250	1088	601	start	M03
626	T104	J34	0	1250	1088	708	start	M03
626	T105	J34	0	1250	1088	582	start	M03
629	T080	J29	4	3125	1125	47	finish	M07
629	T106	J34	0	1250	1088	643	start	M07
629	T107	J34	0	1250	1088	595	start	M07
648	T117	J35	1	937	249	203	submit	-
659	T118	J36	0	312	199	194	submit	-
659	T119	J36	0	312	199	193	submit	-
659	T120	J36	0	312	199	195	submit	-
659	T121	J36	0	312	199	195	submit	-
659	T122	J36	0	312	199	195	submit	-
659	T123	J36	0	312	199	194	submit	-
659	T124	J36	0	312	199	194	submit	-
659	T125	J36	0	312	199	194	submit	-
659	T126	J36	0	312	199	193	submit	-
659	T127	J36	0	312	199	194	submit	-
659	T128	J36	0	312	199	195	submit	-
659	T129	J36	0	312	199	193	submit	-
659	T130	J36	0	312	199	192	submit	-
659	T131	J36	0	312	199	194	submit	-
659	T132	J36	0	312	199	195	submit	-
659	T133	J36	0	312	199	195	submit	-
659	T118	J36	0	312	199	194	start	M01
659	T119	J36	0	312	199	193	start	M02
659	T120	J36	0	312	199	195	start	M02
659	T121	J36	0	312	199	195	start	M06

time	task	job	priority	CPU	memory	run time	event	machine
659	T122	J36	0	312	199	195	start	M07
659	T123	J36	0	312	199	194	start	M07
659	T124	J36	0	312	199	194	start	M08
659	T125	J36	0	312	199	194	start	M08
686	T085	J31	0	937	266	94	finish	M08
686	T087	J31	0	937	266	94	finish	M08
686	T089	J31	0	937	266	94	finish	M08
686	T090	J31	0	937	266	94	finish	M08
686	T117	J35	1	937	249	203	start	M08
686	T108	J34	0	1250	1088	568	start	M08
686	T109	J34	0	1250	1088	636	start	M08
693	T134	J37	9	6250	233	111	submit	-
693	T105	J34	0	1250	1088	582	evict	M03
693	T104	J34	0	1250	1088	708	evict	M03
693	T103	J34	0	1250	1088	601	evict	M03
693	T102	J34	0	1250	1088	571	evict	M03
693	T101	J34	0	1250	1088	750	evict	M03
693	T134	J37	9	6250	233	111	start	M03
694	T135	J38	0	6250	466	45	submit	-
696	T088	J31	0	937	266	104	finish	M08
696	T101	J34	0	1250	1088	750	start	M08
697	T086	J31	0	937	266	105	finish	M08
697	T126	J36	0	312	199	193	start	M08
697	T127	J36	0	312	199	194	start	M08
697	T128	J36	0	312	199	195	start	M08
698	T092	J31	0	937	266	92	finish	M01
698	T129	J36	0	312	199	193	start	M01
698	T130	J36	0	312	199	192	start	M01
701	T084	J31	0	937	266	109	finish	M07
701	T131	J36	0	312	199	194	start	M07
701	T132	J36	0	312	199	195	start	M07
701	T133	J36	0	312	199	195	start	M07
702	T136	J39	4	1250	398	24	submit	-
712	T097	J31	0	937	266	92	finish	M06
713	T091	J31	0	937	266	107	finish	M01
713	T137	J40	8	3125	2386	1883	submit	-
713	T138	J41	8	3125	2386	1517	submit	-
713	T136	J39	4	1250	398	24	start	M01
726	T095	J31	0	937	266	106	finish	M06
726	T098	J31	0	937	266	106	finish	M06
737	T096	J31	0	937	266	117	finish	M06
737	T136	J39	4	1250	398	24	finish	M01
737	T137	J40	8	3125	2386	1883	start	M06
737	T102	J34	0	1250	1088	571	start	M01
742	T093	J31	0	937	266	136	finish	M01
767	T139	J42	4	1874	3180	633	submit	-
767	T140	J42	4	1874	3180	660	submit	-
767	T141	J42	4	1874	3180	659	submit	-
767	T142	J42	4	1874	3180	633	submit	-
780	T143	J43	8	937	1590	718	submit	-
780	T138	J41	8	3125	2386	1517	start	M04
780	T143	J43	8	937	1590	718	start	M01
795	T144	J44	8	1874	466	3292	submit	-
795	T145	J45	8	937	1590	745	submit	-
795	T144	J44	8	1874	466	3292	start	M04
804	T134	J37	9	6250	233	111	finish	M03
804	T145	J45	8	937	1590	745	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
804	T139	J42	4	1874	3180	633	start	M03
804	T103	J34	0	1250	1088	601	start	M03
808	T146	J46	4	625	1288	675	submit	-
808	T147	J46	4	625	1288	668	submit	-
808	T148	J46	4	625	1288	659	submit	-
808	T149	J46	4	625	1288	662	submit	-
823	T094	J31	0	937	266	203	finish	M06
845	T083	J31	0	937	266	253	finish	M07
845	T146	J46	4	625	1288	675	start	M07
852	T119	J36	0	312	199	193	finish	M02
853	T118	J36	0	312	199	194	finish	M01
853	T123	J36	0	312	199	194	finish	M07
853	T124	J36	0	312	199	194	finish	M08
853	T125	J36	0	312	199	194	finish	M08
853	T147	J46	4	625	1288	668	start	M08
854	T120	J36	0	312	199	195	finish	M02
854	T121	J36	0	312	199	195	finish	M06
854	T122	J36	0	312	199	195	finish	M07
854	T140	J42	4	1874	3180	660	start	M06
854	T148	J46	4	625	1288	659	start	M02
854	T149	J46	4	625	1288	662	start	M07
866	T065	J17	9	312	522	628	finish	M02
878	T150	J47	9	6250	1590	66	submit	-
889	T117	J35	1	937	249	203	finish	M08
890	T126	J36	0	312	199	193	finish	M08
890	T130	J36	0	312	199	192	finish	M01
890	T104	J34	0	1250	1088	708	start	M08
891	T127	J36	0	312	199	194	finish	M08
891	T129	J36	0	312	199	193	finish	M01
892	T128	J36	0	312	199	195	finish	M08
895	T131	J36	0	312	199	194	finish	M07
896	T132	J36	0	312	199	195	finish	M07
896	T133	J36	0	312	199	195	finish	M07
926	T151	J48	4	3125	932	1076	submit	-
926	T152	J48	4	3125	932	1056	submit	-
926	T153	J48	4	3125	932	1057	submit	-
926	T154	J48	4	3125	932	1133	submit	-
1020	T155	J49	1	4779	2332	5867	submit	-
1046	T156	J50	0	125	1553	5684	submit	-
1046	T157	J50	0	125	1553	5685	submit	-
1046	T158	J50	0	125	1553	5792	submit	-
1046	T159	J50	0	125	1553	5685	submit	-
1046	T160	J50	0	125	1553	5678	submit	-
1046	T161	J50	0	125	1553	5684	submit	-
1046	T162	J50	0	125	1553	5705	submit	-
1046	T163	J50	0	125	1553	5744	submit	-
1046	T164	J50	0	125	1553	5722	submit	-
1046	T165	J50	0	125	1553	5664	submit	-
1046	T166	J50	0	125	1553	5837	submit	-
1046	T167	J50	0	125	1553	5700	submit	-
1046	T168	J50	0	125	1553	5681	submit	-
1046	T169	J50	0	125	1553	5721	submit	-
1046	T170	J50	0	125	1553	5804	submit	-
1046	T171	J50	0	125	1553	5705	submit	-
1050	T172	J51	0	125	1553	5443	submit	-
1050	T173	J51	0	125	1553	5499	submit	-
1050	T174	J51	0	125	1553	5442	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
1050	T175	J51	0	125	1553	5461	submit	-
1050	T176	J51	0	125	1553	5482	submit	-
1050	T177	J51	0	125	1553	5469	submit	-
1050	T178	J51	0	125	1553	5496	submit	-
1050	T179	J51	0	125	1553	5423	submit	-
1050	T180	J51	0	125	1553	5474	submit	-
1050	T181	J51	0	125	1553	5516	submit	-
1050	T182	J51	0	125	1553	5474	submit	-
1050	T183	J51	0	125	1553	5498	submit	-
1050	T184	J51	0	125	1553	5430	submit	-
1050	T185	J51	0	125	1553	5442	submit	-
1050	T186	J51	0	125	1553	5472	submit	-
1050	T187	J51	0	125	1553	5430	submit	-
1053	T188	J52	0	6250	1244	45	submit	-
1053	T189	J53	0	6250	1244	69	submit	-
1064	T076	J25	8	625	398	621	finish	M01
1072	T190	J54	6	6250	622	114	submit	-
1072	T191	J55	6	625	280	56	submit	-
1072	T192	J56	8	2499	466	1001	submit	-
1072	T191	J55	6	625	280	56	start	M01
1093	T193	J57	1	2499	311	329	submit	-
1128	T191	J55	6	625	280	56	finish	M01
1166	T194	J58	4	3125	777	209	submit	-
1166	T195	J58	4	3125	777	207	submit	-
1166	T196	J58	4	3125	777	206	submit	-
1166	T197	J58	4	3125	777	208	submit	-
1180	T198	J59	4	3125	155	108	submit	-
1180	T199	J59	4	3125	155	86	submit	-
1180	T200	J59	4	3125	155	82	submit	-
1180	T201	J59	4	3125	155	83	submit	-
1192	T202	J60	1	3748	466	215	submit	-
1197	T203	J61	0	1709	267	183	submit	-
1197	T204	J61	0	1709	267	183	submit	-
1197	T205	J61	0	1709	267	164	submit	-
1197	T206	J61	0	1709	267	164	submit	-
1197	T207	J61	0	1709	267	182	submit	-
1197	T208	J61	0	1709	267	184	submit	-
1197	T209	J61	0	1709	267	185	submit	-
1197	T210	J61	0	1709	267	163	submit	-
1197	T211	J61	0	1709	267	183	submit	-
1197	T212	J61	0	1709	267	180	submit	-
1197	T213	J61	0	1709	267	170	submit	-
1197	T214	J61	0	1709	267	166	submit	-
1197	T215	J61	0	1709	267	164	submit	-
1197	T216	J61	0	1709	267	194	submit	-
1197	T217	J61	0	1709	267	174	submit	-
1197	T218	J61	0	1709	267	182	submit	-
1197	T203	J61	0	1709	267	183	start	M03

Cell window 15: Events

Time 187: task T064 (priority 9) evicted task T002 (priority 1) from machine M01.

Time 187: task T064 (priority 9) evicted task T004 (priority 0) from machine M01.

Time 187: task T064 (priority 9) evicted task T043 (priority 1) from machine M01.

Time 187: task T002 restarted on machine M02.

Time 187: task T004 restarted on machine M03.

Time 187: task T043 restarted on machine M05.

Time 385: machine M04 failed; no task was running on it.

Time 693: task T134 (priority 9) evicted task T101 (priority 0) from machine M03.

Time 693: task T134 (priority 9) evicted task T102 (priority 0) from machine M03.

Time 693: task T134 (priority 9) evicted task T103 (priority 0) from machine M03.

Time 693: task T134 (priority 9) evicted task T104 (priority 0) from machine M03.

Time 693: task T134 (priority 9) evicted task T105 (priority 0) from machine M03.

Time 696: task T101 restarted on machine M08.

Time 737: task T102 restarted on machine M01.

Time 780: machine M04 returned.

Time 804: task T103 restarted on machine M03.

Time 890: task T104 restarted on machine M08.

Longest waits (submission to first start, tasks started by the window end): task T040 waited 89 s (submitted at 97, first started at 186); task T140 waited 87 s (submitted at 767, first started at 854); task T108 waited 84 s (submitted at 602, first started at 686); task T109 waited 84 s (submitted at 602, first started at 686); task T039 waited 81 s (submitted at 96, first started at 177).

Cell window 15: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T068	J20	4	1250	870	M01	373	7
T069	J20	4	1250	870	M01	373	5
T102	J34	0	1250	1088	M01	737	108
T143	J43	8	937	1590	M01	780	298
T070	J20	4	1250	870	M02	373	6
T071	J20	4	1250	870	M02	373	20
T077	J26	1	1562	1553	M02	566	266
T081	J30	0	1250	777	M02	590	641
T148	J46	4	625	1288	M02	854	313
T103	J34	0	1250	1088	M03	804	205
T139	J42	4	1874	3180	M03	804	237
T145	J45	8	937	1590	M03	804	349
T203	J61	0	1709	267	M03	1197	180
T138	J41	8	3125	2386	M04	780	1097
T144	J44	8	1874	466	M04	795	2887
T099	J32	8	6250	3821	M05	615	519
T100	J33	0	1250	384	M06	620	273
T137	J40	8	3125	2386	M06	737	1420
T140	J42	4	1874	3180	M06	854	314
T082	J30	0	1250	777	M07	590	488
T106	J34	0	1250	1088	M07	629	72
T107	J34	0	1250	1088	M07	629	24
T146	J46	4	625	1288	M07	845	320
T149	J46	4	625	1288	M07	854	316
T101	J34	0	1250	1088	M08	696	246
T104	J34	0	1250	1088	M08	890	398
T108	J34	0	1250	1088	M08	686	54

task	job	priority	CPU	memory	machine	started	remaining
T109	J34	0	1250	1088	M08	686	122
T147	J46	4	625	1288	M08	853	321

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T150	J47	9	6250	1590	878	66
T192	J56	8	2499	466	1072	1001
T190	J54	6	6250	622	1072	114
T141	J42	4	1874	3180	767	659
T142	J42	4	1874	3180	767	633
T151	J48	4	3125	932	926	1076
T152	J48	4	3125	932	926	1056
T153	J48	4	3125	932	926	1057
T154	J48	4	3125	932	926	1133
T194	J58	4	3125	777	1166	209
T195	J58	4	3125	777	1166	207
T196	J58	4	3125	777	1166	206
T197	J58	4	3125	777	1166	208
T198	J59	4	3125	155	1180	108
T199	J59	4	3125	155	1180	86
T200	J59	4	3125	155	1180	82
T201	J59	4	3125	155	1180	83
T155	J49	1	4779	2332	1020	5867
T193	J57	1	2499	311	1093	329
T202	J60	1	3748	466	1192	215
T105	J34	0	1250	1088	602	582
T110	J34	0	1250	1088	602	590
T111	J34	0	1250	1088	602	849
T112	J34	0	1250	1088	602	601
T113	J34	0	1250	1088	602	761
T114	J34	0	1250	1088	602	619
T115	J34	0	1250	1088	602	601
T116	J34	0	1250	1088	602	571
T135	J38	0	6250	466	694	45
T156	J50	0	125	1553	1046	5684
T157	J50	0	125	1553	1046	5685
T158	J50	0	125	1553	1046	5792
T159	J50	0	125	1553	1046	5685
T160	J50	0	125	1553	1046	5678
T161	J50	0	125	1553	1046	5684
T162	J50	0	125	1553	1046	5705
T163	J50	0	125	1553	1046	5744
T164	J50	0	125	1553	1046	5722
T165	J50	0	125	1553	1046	5664
T166	J50	0	125	1553	1046	5837
T167	J50	0	125	1553	1046	5700
T168	J50	0	125	1553	1046	5681
T169	J50	0	125	1553	1046	5721
T170	J50	0	125	1553	1046	5804
T171	J50	0	125	1553	1046	5705
T172	J51	0	125	1553	1050	5443
T173	J51	0	125	1553	1050	5499
T174	J51	0	125	1553	1050	5442
T175	J51	0	125	1553	1050	5461
T176	J51	0	125	1553	1050	5482
T177	J51	0	125	1553	1050	5469
T178	J51	0	125	1553	1050	5496

task	job	priority	CPU	memory	submitted	run time
T179	J51	0	125	1553	1050	5423
T180	J51	0	125	1553	1050	5474
T181	J51	0	125	1553	1050	5516
T182	J51	0	125	1553	1050	5474
T183	J51	0	125	1553	1050	5498
T184	J51	0	125	1553	1050	5430
T185	J51	0	125	1553	1050	5442
T186	J51	0	125	1553	1050	5472
T187	J51	0	125	1553	1050	5430
T188	J52	0	6250	1244	1053	45
T189	J53	0	6250	1244	1053	69
T204	J61	0	1709	267	1197	183
T205	J61	0	1709	267	1197	164
T206	J61	0	1709	267	1197	164
T207	J61	0	1709	267	1197	182
T208	J61	0	1709	267	1197	184
T209	J61	0	1709	267	1197	185
T210	J61	0	1709	267	1197	163
T211	J61	0	1709	267	1197	183
T212	J61	0	1709	267	1197	180
T213	J61	0	1709	267	1197	170
T214	J61	0	1709	267	1197	166
T215	J61	0	1709	267	1197	164
T216	J61	0	1709	267	1197	194
T217	J61	0	1709	267	1197	174
T218	J61	0	1709	267	1197	182

All machines are up at time 1200.

Cell window 16

Times 0 to 1200. The cluster has 8 machines. 120 tasks of 36 jobs are submitted during the window.

Cell window 16: Machines

machine	CPU	memory	fails at	returns at
M01	6250	3116	-	-
M02	6250	6244	-	-
M03	6250	6244	-	-
M04	6250	1551	324	1321
M05	6250	6244	-	-
M06	6250	3116	-	-
M07	6250	6244	-	-
M08	6250	3116	-	-

Cell window 16: Task log

time	task	job	priority	CPU	memory	run time	event	machine
14	T001	J01	4	3125	777	116	submit	-
14	T002	J01	4	3125	777	113	submit	-
14	T003	J01	4	3125	777	113	submit	-
14	T004	J01	4	3125	777	113	submit	-
14	T001	J01	4	3125	777	116	start	M01
14	T002	J01	4	3125	777	113	start	M01
14	T003	J01	4	3125	777	113	start	M02
14	T004	J01	4	3125	777	113	start	M02
23	T005	J02	1	937	249	203	submit	-
23	T005	J02	1	937	249	203	start	M03
34	T006	J03	0	312	199	194	submit	-
34	T007	J03	0	312	199	193	submit	-
34	T008	J03	0	312	199	195	submit	-
34	T009	J03	0	312	199	195	submit	-
34	T010	J03	0	312	199	195	submit	-
34	T011	J03	0	312	199	194	submit	-
34	T012	J03	0	312	199	194	submit	-
34	T013	J03	0	312	199	194	submit	-
34	T014	J03	0	312	199	193	submit	-
34	T015	J03	0	312	199	194	submit	-
34	T016	J03	0	312	199	195	submit	-
34	T017	J03	0	312	199	193	submit	-
34	T018	J03	0	312	199	192	submit	-
34	T019	J03	0	312	199	194	submit	-
34	T020	J03	0	312	199	195	submit	-
34	T021	J03	0	312	199	195	submit	-
34	T006	J03	0	312	199	194	start	M03
34	T007	J03	0	312	199	193	start	M03
34	T008	J03	0	312	199	195	start	M03
34	T009	J03	0	312	199	195	start	M03
34	T010	J03	0	312	199	195	start	M03
34	T011	J03	0	312	199	194	start	M03
34	T012	J03	0	312	199	194	start	M03
34	T013	J03	0	312	199	194	start	M03
34	T014	J03	0	312	199	193	start	M03
34	T015	J03	0	312	199	194	start	M03
34	T016	J03	0	312	199	195	start	M03
34	T017	J03	0	312	199	193	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
34	T018	J03	0	312	199	192	start	M03
34	T019	J03	0	312	199	194	start	M03
34	T020	J03	0	312	199	195	start	M03
34	T021	J03	0	312	199	195	start	M03
68	T022	J04	9	6250	233	111	submit	-
68	T022	J04	9	6250	233	111	start	M04
69	T023	J05	0	6250	466	45	submit	-
69	T023	J05	0	6250	466	45	start	M05
88	T024	J06	8	3125	2386	1616	submit	-
88	T024	J06	8	3125	2386	1616	start	M06
114	T023	J05	0	6250	466	45	finish	M05
116	T025	J07	4	1250	466	194	submit	-
116	T026	J07	4	1250	466	190	submit	-
116	T027	J07	4	1250	466	193	submit	-
116	T028	J07	4	1250	466	191	submit	-
116	T025	J07	4	1250	466	194	start	M05
116	T026	J07	4	1250	466	190	start	M05
116	T027	J07	4	1250	466	193	start	M05
116	T028	J07	4	1250	466	191	start	M05
127	T002	J01	4	3125	777	113	finish	M01
127	T003	J01	4	3125	777	113	finish	M02
127	T004	J01	4	3125	777	113	finish	M02
130	T001	J01	4	3125	777	116	finish	M01
141	T029	J08	8	937	1590	979	submit	-
141	T029	J08	8	937	1590	979	start	M01
155	T030	J09	8	937	1590	718	submit	-
155	T030	J09	8	937	1590	718	start	M02
170	T031	J10	8	937	1590	688	submit	-
170	T031	J10	8	937	1590	688	start	M02
179	T022	J04	9	6250	233	111	finish	M04
183	T032	J11	4	625	1288	675	submit	-
183	T033	J11	4	625	1288	668	submit	-
183	T034	J11	4	625	1288	659	submit	-
183	T035	J11	4	625	1288	662	submit	-
183	T032	J11	4	625	1288	675	start	M01
183	T033	J11	4	625	1288	668	start	M02
183	T034	J11	4	625	1288	659	start	M02
183	T035	J11	4	625	1288	662	start	M04
226	T005	J02	1	937	249	203	finish	M03
226	T018	J03	0	312	199	192	finish	M03
227	T007	J03	0	312	199	193	finish	M03
227	T014	J03	0	312	199	193	finish	M03
227	T017	J03	0	312	199	193	finish	M03
228	T006	J03	0	312	199	194	finish	M03
228	T011	J03	0	312	199	194	finish	M03
228	T012	J03	0	312	199	194	finish	M03
228	T013	J03	0	312	199	194	finish	M03
228	T015	J03	0	312	199	194	finish	M03
228	T019	J03	0	312	199	194	finish	M03
229	T008	J03	0	312	199	195	finish	M03
229	T009	J03	0	312	199	195	finish	M03
229	T010	J03	0	312	199	195	finish	M03
229	T016	J03	0	312	199	195	finish	M03
229	T020	J03	0	312	199	195	finish	M03
229	T021	J03	0	312	199	195	finish	M03
253	T036	J12	9	6250	1590	66	submit	-
253	T036	J12	9	6250	1590	66	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
306	T026	J07	4	1250	466	190	finish	M05
307	T028	J07	4	1250	466	191	finish	M05
309	T027	J07	4	1250	466	193	finish	M05
310	T025	J07	4	1250	466	194	finish	M05
319	T036	J12	9	6250	1590	66	finish	M03
324	T035	J11	4	625	1288	662	fail	M04
324	T035	J11	4	625	1288	662	start	M03
395	T037	J13	1	4779	2332	5867	submit	-
395	T037	J13	1	4779	2332	5867	start	M03
425	T038	J14	0	125	1553	5443	submit	-
425	T039	J14	0	125	1553	5499	submit	-
425	T040	J14	0	125	1553	5442	submit	-
425	T041	J14	0	125	1553	5461	submit	-
425	T042	J14	0	125	1553	5482	submit	-
425	T043	J14	0	125	1553	5469	submit	-
425	T044	J14	0	125	1553	5496	submit	-
425	T045	J14	0	125	1553	5423	submit	-
425	T046	J14	0	125	1553	5474	submit	-
425	T047	J14	0	125	1553	5516	submit	-
425	T048	J14	0	125	1553	5474	submit	-
425	T049	J14	0	125	1553	5498	submit	-
425	T050	J14	0	125	1553	5430	submit	-
425	T051	J14	0	125	1553	5442	submit	-
425	T052	J14	0	125	1553	5472	submit	-
425	T053	J14	0	125	1553	5430	submit	-
425	T038	J14	0	125	1553	5443	start	M03
425	T039	J14	0	125	1553	5499	start	M05
425	T040	J14	0	125	1553	5442	start	M05
425	T041	J14	0	125	1553	5461	start	M05
425	T042	J14	0	125	1553	5482	start	M05
425	T043	J14	0	125	1553	5469	start	M07
425	T044	J14	0	125	1553	5496	start	M07
425	T045	J14	0	125	1553	5423	start	M07
425	T046	J14	0	125	1553	5474	start	M07
425	T047	J14	0	125	1553	5516	start	M08
425	T048	J14	0	125	1553	5474	start	M08
447	T054	J15	8	4626	466	3837	submit	-
464	T055	J16	4	312	448	1060	submit	-
464	T055	J16	4	312	448	1060	start	M02
468	T056	J17	1	2499	311	329	submit	-
468	T056	J17	1	2499	311	329	start	M06
572	T057	J18	0	1709	267	183	submit	-
572	T058	J18	0	1709	267	183	submit	-
572	T059	J18	0	1709	267	164	submit	-
572	T060	J18	0	1709	267	164	submit	-
572	T061	J18	0	1709	267	182	submit	-
572	T062	J18	0	1709	267	184	submit	-
572	T063	J18	0	1709	267	185	submit	-
572	T064	J18	0	1709	267	163	submit	-
572	T065	J18	0	1709	267	183	submit	-
572	T066	J18	0	1709	267	180	submit	-
572	T067	J18	0	1709	267	170	submit	-
572	T068	J18	0	1709	267	166	submit	-
572	T069	J18	0	1709	267	164	submit	-
572	T070	J18	0	1709	267	194	submit	-
572	T071	J18	0	1709	267	174	submit	-
572	T072	J18	0	1709	267	182	submit	-

time	task	job	priority	CPU	memory	run time	event	machine
608	T073	J19	0	3125	388	272	submit	-
622	T074	J20	6	1250	612	245	submit	-
622	T075	J20	6	1250	612	265	submit	-
622	T076	J20	6	1250	612	252	submit	-
622	T077	J20	6	1250	612	221	submit	-
622	T078	J21	6	937	221	74	submit	-
622	T079	J21	6	937	221	82	submit	-
622	T080	J21	6	937	221	56	submit	-
622	T081	J21	6	937	221	113	submit	-
622	T078	J21	6	937	221	74	start	M01
667	T082	J22	9	6250	3180	200	submit	-
667	T042	J14	0	125	1553	5482	evict	M05
667	T041	J14	0	125	1553	5461	evict	M05
667	T040	J14	0	125	1553	5442	evict	M05
667	T039	J14	0	125	1553	5499	evict	M05
667	T082	J22	9	6250	3180	200	start	M05
669	T083	J23	0	6250	466	37	submit	-
670	T084	J24	1	1250	1590	384	submit	-
696	T078	J21	6	937	221	74	finish	M01
696	T079	J21	6	937	221	82	start	M01
703	T085	J25	1	1250	466	187	submit	-
705	T086	J26	0	625	252	160	submit	-
705	T087	J26	0	625	252	171	submit	-
705	T088	J26	0	625	252	169	submit	-
705	T089	J26	0	625	252	158	submit	-
705	T090	J26	0	625	252	160	submit	-
705	T091	J26	0	625	252	157	submit	-
705	T092	J26	0	625	252	162	submit	-
705	T093	J26	0	625	252	162	submit	-
705	T094	J26	0	625	252	160	submit	-
705	T095	J26	0	625	252	160	submit	-
705	T096	J26	0	625	252	171	submit	-
705	T097	J26	0	625	252	159	submit	-
705	T098	J26	0	625	252	161	submit	-
705	T099	J26	0	625	252	172	submit	-
705	T100	J26	0	625	252	161	submit	-
705	T101	J26	0	625	252	159	submit	-
705	T086	J26	0	625	252	160	start	M03
705	T087	J26	0	625	252	171	start	M06
778	T079	J21	6	937	221	82	finish	M01
778	T080	J21	6	937	221	56	start	M01
797	T056	J17	1	2499	311	329	finish	M06
797	T081	J21	6	937	221	113	start	M06
797	T088	J26	0	625	252	169	start	M06
834	T080	J21	6	937	221	56	finish	M01
842	T034	J11	4	625	1288	659	finish	M02
842	T074	J20	6	1250	612	245	start	M02
842	T075	J20	6	1250	612	265	start	M02
851	T033	J11	4	625	1288	668	finish	M02
851	T076	J20	6	1250	612	252	start	M02
858	T031	J10	8	937	1590	688	finish	M02
858	T032	J11	4	625	1288	675	finish	M01
858	T054	J15	8	4626	466	3837	start	M01
858	T077	J20	6	1250	612	221	start	M02
858	T089	J26	0	625	252	158	start	M01
865	T086	J26	0	625	252	160	finish	M03
865	T090	J26	0	625	252	160	start	M03

time	task	job	priority	CPU	memory	run time	event	machine
867	T082	J22	9	6250	3180	200	finish	M05
867	T084	J24	1	1250	1590	384	start	M05
867	T085	J25	1	1250	466	187	start	M05
867	T039	J14	0	125	1553	5499	start	M05
867	T040	J14	0	125	1553	5442	start	M05
867	T057	J18	0	1709	267	183	start	M05
867	T058	J18	0	1709	267	183	start	M05
873	T030	J09	8	937	1590	718	finish	M02
873	T041	J14	0	125	1553	5461	start	M02
873	T042	J14	0	125	1553	5482	start	M02
876	T087	J26	0	625	252	171	finish	M06
876	T091	J26	0	625	252	157	start	M06
889	T102	J27	6	937	1244	318	submit	-
910	T081	J21	6	937	221	113	finish	M06
931	T103	J28	6	625	280	105	submit	-
966	T088	J26	0	625	252	169	finish	M06
966	T103	J28	6	625	280	105	start	M06
967	T104	J29	9	6250	233	100	submit	-
967	T058	J18	0	1709	267	183	evict	M05
967	T057	J18	0	1709	267	183	evict	M05
967	T040	J14	0	125	1553	5442	evict	M05
967	T039	J14	0	125	1553	5499	evict	M05
967	T085	J25	1	1250	466	187	evict	M05
967	T084	J24	1	1250	1590	384	evict	M05
967	T104	J29	9	6250	233	100	start	M05
974	T105	J30	9	6250	1590	70	submit	-
974	T046	J14	0	125	1553	5474	evict	M07
974	T045	J14	0	125	1553	5423	evict	M07
974	T044	J14	0	125	1553	5496	evict	M07
974	T043	J14	0	125	1553	5469	evict	M07
974	T105	J30	9	6250	1590	70	start	M07
976	T106	J31	4	3125	448	30	submit	-
986	T035	J11	4	625	1288	662	finish	M03
989	T107	J32	8	3125	2386	268	submit	-
989	T039	J14	0	125	1553	5499	start	M03
1004	T108	J33	0	1250	622	50	submit	-
1004	T109	J33	0	1250	622	46	submit	-
1004	T110	J33	0	1250	622	47	submit	-
1004	T111	J33	0	1250	622	50	submit	-
1004	T112	J33	0	1250	622	47	submit	-
1004	T113	J33	0	1250	622	50	submit	-
1004	T114	J33	0	1250	622	48	submit	-
1016	T089	J26	0	625	252	158	finish	M01
1016	T092	J26	0	625	252	162	start	M01
1025	T090	J26	0	625	252	160	finish	M03
1025	T093	J26	0	625	252	162	start	M03
1030	T115	J34	8	1874	3180	309	submit	-
1033	T091	J26	0	625	252	157	finish	M06
1033	T057	J18	0	1709	267	183	start	M06
1042	T116	J35	6	625	280	247	submit	-
1044	T105	J30	9	6250	1590	70	finish	M07
1044	T107	J32	8	3125	2386	268	start	M07
1044	T115	J34	8	1874	3180	309	start	M07
1044	T116	J35	6	625	280	247	start	M07
1044	T094	J26	0	625	252	160	start	M07
1067	T104	J29	9	6250	233	100	finish	M05
1067	T102	J27	6	937	1244	318	start	M05

time	task	job	priority	CPU	memory	run time	event	machine
1067	T106	J31	4	3125	448	30	start	M05
1067	T084	J24	1	1250	1590	384	start	M05
1067	T040	J14	0	125	1553	5442	start	M05
1067	T095	J26	0	625	252	160	start	M05
1071	T103	J28	6	625	280	105	finish	M06
1071	T096	J26	0	625	252	171	start	M06
1073	T117	J36	6	937	230	141	submit	-
1073	T118	J36	6	937	230	46	submit	-
1073	T119	J36	6	937	230	55	submit	-
1073	T120	J36	6	937	230	59	submit	-
1079	T077	J20	6	1250	612	221	finish	M02
1079	T117	J36	6	937	230	141	start	M02
1079	T118	J36	6	937	230	46	start	M02
1087	T074	J20	6	1250	612	245	finish	M02
1087	T119	J36	6	937	230	55	start	M02
1097	T106	J31	4	3125	448	30	finish	M05
1097	T120	J36	6	937	230	59	start	M05
1097	T085	J25	1	1250	466	187	start	M05
1097	T097	J26	0	625	252	159	start	M05
1103	T076	J20	6	1250	612	252	finish	M02
1103	T098	J26	0	625	252	161	start	M02
1107	T075	J20	6	1250	612	265	finish	M02
1107	T043	J14	0	125	1553	5469	start	M02
1120	T029	J08	8	937	1590	979	finish	M01
1120	T044	J14	0	125	1553	5496	start	M01
1120	T099	J26	0	625	252	172	start	M01
1125	T118	J36	6	937	230	46	finish	M02
1125	T058	J18	0	1709	267	183	start	M02
1142	T119	J36	6	937	230	55	finish	M02
1142	T059	J18	0	1709	267	164	start	M02
1156	T120	J36	6	937	230	59	finish	M05
1156	T100	J26	0	625	252	161	start	M05
1178	T092	J26	0	625	252	162	finish	M01
1178	T101	J26	0	625	252	159	start	M01
1187	T093	J26	0	625	252	162	finish	M03

Cell window 16: Events

Time 324: machine M04 failed; task T035 stopped and went back to the queue.

Time 324: task T035 restarted on machine M03.

Time 667: task T082 (priority 9) evicted task T039 (priority 0) from machine M05.

Time 667: task T082 (priority 9) evicted task T040 (priority 0) from machine M05.

Time 667: task T082 (priority 9) evicted task T041 (priority 0) from machine M05.

Time 667: task T082 (priority 9) evicted task T042 (priority 0) from machine M05.

Time 867: task T039 restarted on machine M05.

Time 867: task T040 restarted on machine M05.

Time 873: task T041 restarted on machine M02.

Time 873: task T042 restarted on machine M02.

Time 967: task T104 (priority 9) evicted task T039 (priority 0) from machine M05.

Time 967: task T104 (priority 9) evicted task T040 (priority 0) from machine M05.

Time 967: task T104 (priority 9) evicted task T057 (priority 0) from machine M05.

Time 967: task T104 (priority 9) evicted task T058 (priority 0) from machine M05.

Time 967: task T104 (priority 9) evicted task T084 (priority 1) from machine M05.

Time 967: task T104 (priority 9) evicted task T085 (priority 1) from machine M05.

Time 974: task T105 (priority 9) evicted task T043 (priority 0) from machine M07.

Time 974: task T105 (priority 9) evicted task T044 (priority 0) from machine M07.

Time 974: task T105 (priority 9) evicted task T045 (priority 0) from machine M07.

Time 974: task T105 (priority 9) evicted task T046 (priority 0) from machine M07.

Time 989: task T039 restarted on machine M03.

Time 1033: task T057 restarted on machine M06.

Time 1067: task T040 restarted on machine M05.

Time 1067: task T084 restarted on machine M05.

Time 1097: task T085 restarted on machine M05.

Time 1107: task T043 restarted on machine M02.

Time 1120: task T044 restarted on machine M01.

Time 1125: task T058 restarted on machine M02.

Longest waits (submission to first start, tasks started by the window end): task T059 waited 570 s (submitted at 572, first started at 1142); task T101 waited 473 s (submitted at 705, first started at 1178); task T100 waited 451 s (submitted at 705, first started at 1156); task T099 waited 415 s (submitted at 705, first started at 1120); task T054 waited 411 s (submitted at 447, first started at 858).

Cell window 16: Queue at window end

State after the events at time 1200.

Tasks running at time 1200 (remaining run time = seconds still to run after time 1200):

task	job	priority	CPU	memory	machine	started	remaining
T044	J14	0	125	1553	M01	1120	5416
T054	J15	8	4626	466	M01	858	3495
T099	J26	0	625	252	M01	1120	92
T101	J26	0	625	252	M01	1178	137
T041	J14	0	125	1553	M02	873	5134
T042	J14	0	125	1553	M02	873	5155
T043	J14	0	125	1553	M02	1107	5376
T055	J16	4	312	448	M02	464	324
T058	J18	0	1709	267	M02	1125	108
T059	J18	0	1709	267	M02	1142	106
T098	J26	0	625	252	M02	1103	64
T117	J36	6	937	230	M02	1079	20
T037	J13	1	4779	2332	M03	395	5062
T038	J14	0	125	1553	M03	425	4668
T039	J14	0	125	1553	M03	989	5288
T040	J14	0	125	1553	M05	1067	5309
T084	J24	1	1250	1590	M05	1067	251
T085	J25	1	1250	466	M05	1097	84
T095	J26	0	625	252	M05	1067	27
T097	J26	0	625	252	M05	1097	56
T100	J26	0	625	252	M05	1156	117
T102	J27	6	937	1244	M05	1067	185

task	job	priority	CPU	memory	machine	started	remaining
T024	J06	8	3125	2386	M06	88	504
T057	J18	0	1709	267	M06	1033	16
T096	J26	0	625	252	M06	1071	42
T094	J26	0	625	252	M07	1044	4
T107	J32	8	3125	2386	M07	1044	112
T115	J34	8	1874	3180	M07	1044	153
T116	J35	6	625	280	M07	1044	91
T047	J14	0	125	1553	M08	425	4741
T048	J14	0	125	1553	M08	425	4699

Tasks waiting at time 1200, in queue order (each will run its full run time once started):

task	job	priority	CPU	memory	submitted	run time
T045	J14	0	125	1553	425	5423
T046	J14	0	125	1553	425	5474
T049	J14	0	125	1553	425	5498
T050	J14	0	125	1553	425	5430
T051	J14	0	125	1553	425	5442
T052	J14	0	125	1553	425	5472
T053	J14	0	125	1553	425	5430
T060	J18	0	1709	267	572	164
T061	J18	0	1709	267	572	182
T062	J18	0	1709	267	572	184
T063	J18	0	1709	267	572	185
T064	J18	0	1709	267	572	163
T065	J18	0	1709	267	572	183
T066	J18	0	1709	267	572	180
T067	J18	0	1709	267	572	170
T068	J18	0	1709	267	572	166
T069	J18	0	1709	267	572	164
T070	J18	0	1709	267	572	194
T071	J18	0	1709	267	572	174
T072	J18	0	1709	267	572	182
T073	J19	0	3125	388	608	272
T083	J23	0	6250	466	669	37
T108	J33	0	1250	622	1004	50
T109	J33	0	1250	622	1004	46
T110	J33	0	1250	622	1004	47
T111	J33	0	1250	622	1004	50
T112	J33	0	1250	622	1004	47
T113	J33	0	1250	622	1004	50
T114	J33	0	1250	622	1004	48

Machine M04 is down at time 1200; it returns at time 1321.